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Asynchronous Barter, Periodic Reconciliation: Evidence for a Scale-Free Commercial Network in the Indus Valley Civilization

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> Yet to the steadfast eye of one standing upon the shore of things, looking chiefly and most lovingly upon that side on which the sun shines and that we feel joyously to be life, the heart is ever gladdened by the beauty, the exquisite spontaneity, with which life seeks and takes on its forms in an accord perfectly responsive to its needs. It seems ever as though the life and the form were absolutely one and inseparable, so adequate is the sense of fulfilment.
> Whether it be the sweeping eagle in his flight, or the open appleblossom, the toiling work-horse, the blithe swan, the branching oak, the winding stream at its base, the drifting clouds, over all the coursing sun, —*form ever follows function*—, and this is the law. Where function does not change, form does not change. The granite rocks, the everbrooding hills, remain for ages; the lightning lives, comes into shape, and dies, in a twinkling.
> It is the pervading law of all things organic and inorganic, of all things physical and metaphysical, of all things human and all things superhuman, of all true manifestations of the head, of the heart, of the soul, that the life is recognizable in its expression, that form ever follows function.—*This is the law*—.
>
> Louis H. Sullivan, *The Tall Office Building Artistically Considered*

Abstract

The Harappan Civilization (c. 2600–1900 BCE) achieved extraordinary standardization across a vast region without clear evidence of centralized rule. We propose that it was a merchant commonwealth organized as a scale-free commercial network, with AP2PBS (Asynchronous Peer-to-Peer Barter Settlement) and the Unicorn Logistics

Confederation (ULC) serving as inferential model labels for its coordinating institutions.

Four lines of evidence support this interpretation. (1) **Statistical**: Unicorn with $\alpha \approx 1.80$, and offering stand frequencies follow a power-law distribution with $\alpha \approx 2.3$–2.6, while seal counts across sites follow $\alpha \approx 1.55$; all fit heavy-tailed organization better than exponential alternatives. (2) **Attribute correlations**: non-random associations between offering stands and anatomical attributes, including Cramér's V patterns across market tiers, are consistent with deliberate institutional design of visual confirmation bundles. (3) **Archaeological**: seal evolution, contextual distributions, and warehouse logistics support the operational requirements of long-distance asynchronous exchange. (4) **Architectural**: Mohenjo-daro's halls, bath, warehouses, and cart-compatible gridiron are consistent with infrastructure for commerce and coordination.

Taken together, these findings are consistent with AP2PBS and ULC as models of how independent participants could standardize authentication, exchange value asynchronously, and coordinate across distance without a hierarchical state. Scale-free topology also offers a testable explanation for both Harappan longevity and its vulnerability to hub failure and cascading decline. The argument is inferential: the archaeological record does not directly identify AP2PBS or ULC by name, but its statistical, material, and architectural patterns are compatible with this decentralized commercial model.

1 Introduction

The Harappan Civilization (also known as the Indus Valley Civilization) flourished across a vast region of South Asia from approximately 2600 to 1900 BCE, spanning over a million square kilometers across diverse ecological zones. The civilization is renowned for its remarkable standardization: gridiron-planned cities, uniform brick ratios, calibrated weights, and coordinated drainage systems. Yet the archaeological record yields no palaces, no royal tombs, no temples, no monumental inscriptions glorifying rulers, and no obvious centralized administrative authority. This presents a fundamental paradox: How did a civilization encompassing such vast territory achieve and maintain standardization without the hierarchical apparatus (kings, priests, bureaucrats, armies) that characterizes other Bronze Age states?

Among the civilization's most distinctive artifacts are steatite seals, typically measuring 2–3 cm per side, bearing finely carved iconographic motifs and an undeciphered script. Over 3,700 seals are known from Harappan sites. Yet the accompanying material record presents a striking contrast to other ancient civilizations. Mesopotamian systems generated abundant clay sealings on packages and administrative records—evidence of transaction documentation and centralized archiving. By contrast, the Harappan record contains approximately 200 tablets and a similar number of sealings. This scarcity is not a preservation accident: as Frenez (2024) documents, the majority of known Harappan sealings come from a single deposit at Lothal, preserved only due to accidental fire. Most sites yield no sealings at all. This inverts the expected pattern: abundant seals (identity tokens) but scarce sealings and tablets (transaction records). If Harappan seals functioned like Mesopotamian seals (authenticating individual transactions), we should expect abundant sealings corresponding to each transaction. Instead, we find the opposite. This raises a fundamental question: What system would generate abundant seals but systematically scarce (or absent except by

accident) sealings and tablets?

Additional puzzles compound this enigma. Monumental structures at Mohenjo-daro—the Great Hall, Pillared Hall, and Great Bath—lack parallels to temples, palaces, or administrative centers known from other civilizations. Their function remains unclear. The most common seal motif is the unicorn, a wholly invented animal with no natural prototype, appearing suddenly and dominating approximately 60% of mature seals and then disappearing as suddenly after the Mature Harappan period. The offering stand—a non-animal element on the seal—exhibits far more variation than any other attribute; it has been variously interpreted. Kenoyer (2005) documents that "growth of cities such as Harappa appears to be directly linked to increase in regional and long distance trade." He also notes that during the Mature Harappan period ox carts were used in Harappan streets, where major streets allowed two-way traffic while narrow gates were used at city gates, and that "this feature appears to be distinct of Indus cities and has not been documented in early urban centers of Mesopotamia or Egypt". Material culture evidence shows toy carts and boats increase precisely when urban planning transitions from organic to gridded layouts. Why would a logistics innovation (the cart) correlate with urbanization? These patterns suggest an underlying organizing principle distinct from state administrative models.

This paper investigates whether quantitative analysis of seal distributions and archaeological evidence can clarify these puzzles. We employ network science, statistical analysis of frequency distributions, and comparative archaeological evidence to test whether the Harappan seal system encoded commercial organization rather than reflecting primarily priest-, ruler-, or king-based administrative models proposed in earlier interpretations.

Throughout this paper, the term “guild” is used heuristically to denote a recurring commercial affiliation or trading entity represented within the seal system. Given the archaeological contexts of seals and sealings—including workshops, gateways, storage areas, jars, and packages—the term is used simply as a convenient label for economically meaningful groups participating in exchange networks.

2 Assumptions, Frameworks and Mechanisms

Our analysis is based on assumptions grounded in archaeological evidence and employs two complementary theoretical frameworks. First, network externalities explain how independent merchants converge on shared standards through self-reinforcing adoption without centralized enforcement. Second, power law mechanisms (Barabási-Albert and Bentley-Hahn-Shennan models) characterize the statistical signatures that distinguish between preferential attachment (market-driven selection) and neutral drift (random cultural transmission). Together, these frameworks allow us to test whether Harappan seals exhibit patterns consistent with a merchant commonwealth organized by network effects, or alternative models of social organization.

2.1 Core Assumptions

This analysis rests on some assumptions grounded in archaeological evidence. First, the Harappan civilization operated as a merchant and trade network, not primarily as a state

administrative hierarchy. As Kenoyer (2005) documents, "growth of cities such as Harappa appears to be directly linked to increase in regional and long distance trade." The material record confirms this: ox carts, gridded streets, and logistics infrastructure appear unique to Indus cities, and toy carts and boats proliferate precisely when urban planning transitions to gridiron layouts.

Second, seals functioned in commercial and logistic contexts, not in ceremonial or religious roles. Mukhopadhyay (2023) demonstrates that "inscribed seals concentrate in functionally specific contexts," including city gates (traffic control), craft workshops (production control), and storage/warehouse systems (access control). Sealings appear "attached to storage containers and warehouse locking systems," often alongside "standardized Indus weights." Frenez (2024) documents 93 sealings from Lothal's warehouse excavation, with dense clusters reaching 70 sealings in a single square meter, confirming that seals authenticated goods in exchange contexts.

Third, seal distributions and attribute patterns reflect the underlying structure of the commercial network, not random artisanal variation or ceremonial convention. This assumption—that quantitative patterns in material culture encode organizational logic—follows network science approaches to archaeological data and is tested empirically against the data presented in this study.

And finally, we consider seals as a technology that enabled trade - initially it was local, then regional and finally pan-Harappan.

Even though trade is considered diagnostic for Harappan civilization, how it was organized remains under-theorized. We assume that there must have been a method by which the USS functioned as a trade and logistics system for inter-regional trade. It enabled merchants to ship goods securely across geographically dispersed, linguistically diverse networks, where packages could be verified at origin and claimed reliably at destination by trading partners with perhaps no shared language or literacy. This contrasts with religious, administrative, socio-political or ceremonial seal functions.

2.2 Network Externalities

The standardization patterns documented in this study suggest that the Sindhu template—the gridiron-planned urban infrastructure, bathing and drainage systems, standardized brick ratios, and calibrated weights system—and the unicorn seal system exhibit classic network externalities, where the value of adopting a standard increases with the number of other adopters. In the framework of Katz & Shapiro (1985), once a critical mass of merchants adopted Harappan weights and measures, defection became individually irrational regardless of whether any central authority enforced compliance; the standard sustained itself through switching costs and lock-in. Our proposed unicorn seal system exhibits a complementary but distinct dynamic: the seal functions as a two-layer signaling system, where the iconographic layer provides cross-linguistic routing cues across what was likely a multilingual trade zone, while the script layer encodes transaction-level markers between commercial principals. This composite function—simultaneously a network recognition token and a compressed transaction record—is consistent with what Arthur (1989) predicts as the outcome of competing conventions under increasing returns. In that process, the iconographic convention that maximized cross-network legibility accumulated transactional volume, deepened

counterparty recognition, and further entrenched the convention, producing the preferential attachment dynamics documented in the present study without requiring centralized enforcement.

2.3 Power Law Mechanisms: Two Models

Power law distributions are characterized by the probability density function:

$$P(k) \propto k^{-\alpha}$$

where k is a quantity (e.g., frequency, size, degree) and α is the power law exponent. This mathematical form produces the characteristic shape of scale-free distributions: a few entities of very large size, followed by a long tail of progressively smaller entities.

Power laws can arise from multiple distinct generative mechanisms. In cultural and archaeological contexts, distinguishing between mechanisms provides insight into the underlying social and economic dynamics. We focus on two mechanisms particularly relevant to Harappan seals.

2.3.1 Mechanism 1: Barabási-Albert (BA) Preferential Attachment

The Barabási-Albert (BA) model (Barabási & Albert, 1999) describes growth driven by preferential attachment: new nodes joining a network form connections to existing nodes with probability proportional to the nodes' existing degree, creating a self-reinforcing "rich get richer" dynamic. The pure BA model produces a power law with exponent $\alpha = 3$. Empirical examples with confirmed preferential attachment mechanism include scientific citation networks, internet router topology, and actor collaboration networks (Jeong et al., 2003). For Harappan seals, the BA model predicts that successful commercial entities gain more links over time as new merchants preferentially adopt commercial connections to established, reputable identities.

However, the Bianconi–Barabási (BB) fitness model (Bianconi & Barabási, 2001) extends BA by allowing nodes to have intrinsic heterogeneous "fitness" values that modulate their competitiveness in preferential attachment. Under BB dynamics, a node with high fitness attracts links more readily than degree alone would predict, while a node with low fitness attracts links more slowly. The BB model predicts power law exponents in the range $2 < \alpha < 3$ depending on the distribution of fitness values. For Harappan seals, BB is appropriate if commercial entities possessed distinct reputational advantages (fitness) — such as commodity specialization, regional reach, or guild affiliation — that influenced their appeal to new traders independently of their current network position.

2.3.2 Mechanism 2: Bentley-Hahn-Shennan (BHS) Neutral Drift / Random Copying

The Bentley-Hahn-Shennan (BHS) model (Bentley et al., 2004) describes cultural transmission through unbiased (random) copying without selective preference. Under pure neutrality, variants drift randomly in frequency, with exponent typically $1.5 < \alpha < 2$. Empirical examples include baby names in U.S. birth records (Bentley et al., 2004), pottery vessel forms

among potters without functional pressure, and artisanal variation where minor differences accumulate through copying mistakes and individual preference.

A critical point: Bentley et al. explicitly identify that **non-random correlations between attributes constitute deviation from neutrality**. They note that while many pottery motifs followed neutral drift, "specific pottery motifs, perhaps representing prominent households, were not neutral."

3 Methods

We fit power law models using the `powerlaw` Python package (Alstott et al., 2014), which implements the framework of Clauset et al. (2009). Exponents (α) were estimated by maximum likelihood estimation (MLE), and the lower bound $x_{\min}$ was selected by minimizing the Kolmogorov-Smirnov (KS) distance. We also compared power law fits against alternative distributions (exponential, lognormal, stretched exponential, and truncated power law) using log-likelihood ratio tests.

Archaeological samples are sparse: the number of distinct types per attribute, or of sites and motifs, is typically in the tens, well short of the $n > 100$ tail observations that Clauset et al. (2009) recommend for precise parameter estimation, and more data cannot be recovered. We therefore do not treat α as a precisely estimated constant. Three converging criteria are used instead: (i) a bootstrap goodness-of-fit test, which asks whether the power law is a plausible generative model for the observed distribution; (ii) a linear trend on the log–log rank-frequency plot; and (iii) rejection of the exponential, the natural null for a system without preferential attachment, by its own bootstrap goodness-of-fit test as well as by likelihood ratio. Where lognormal and truncated power law cannot be distinguished from the power law, as expected with small samples, we say so. Results are reported as consistent with scale-free structure, not as proof of a particular generative mechanism.

To assess attribute co-occurrence, we computed Cramér's V for associations between offering stand type and each of the ten anatomical attributes (ear, halter, neck, eye, head, hoof, tail, pizzle, leg, horn). Statistical significance was evaluated with χ^2 tests and Bonferroni correction for multiple comparisons. Because these correlation tests require both a valid stand value and a valid attribute value, they use pairwise deletion: seals coded as 'n/a' or 'blank' for a given field are excluded only from that specific test. Accordingly, the effective sample size is lower than the full 500-seal corpus and varies by attribute.

4 Harappan Seal Analysis

We use the seal corpus catalogued by Ameri (2013), which records stamp seals from six sites: Mohenjo-daro, Harappa, Chanhu-daro, Lothal, Kalibangan, and Banawali. The corpus holds $N = 1{,}048$ motif-classified seals in $n = 23$ iconographic motif classes.

The distribution is highly skewed. The Unicorn alone accounts for 800 seals ($\approx 76.3\%$); the other 22 animal, human, and composite motifs share the remaining 23.7%, most with single-digit counts. We fit a power law ($\mathcal{P}(x) \propto x^{-\alpha}$) to the per-motif totals by maximum likelihood, with $x_{\min}$ chosen by minimizing the Kolmogorov–Smirnov (KS) distance (Clauset

et al., 2009; Alstott et al., 2014). Goodness of fit is assessed by a 2,500-draw parametric bootstrap. Results appear in Tables 1 and 2 and Figure 1.

Table 1: Power-Law Statistical Fit Parameters for Indus Seal Motifs.

Parameter	**Value**
Motif types	23
Total seals	1048
α	1.7988 ($\pm$0.1883)
$x_{\min}$	4
Motif types with $x \geq x_{\min}$	18
KS distance D	0.0678
Goodness-of-fit p-value	0.9584

Table 2: Power law versus alternative distributions (loglikelihood ratio R; $R > 0$ favors the power law), and the exponential goodness-of-fit bootstrap.

Alternative	R	p
Exponential	22.8914	0.0050
Lognormal	-0.0070	0.8421
Truncated power law	-0.0190	0.8455
Exponential GoF: $D = 0.5931$, $p < 0.0004$ (0/2500)		

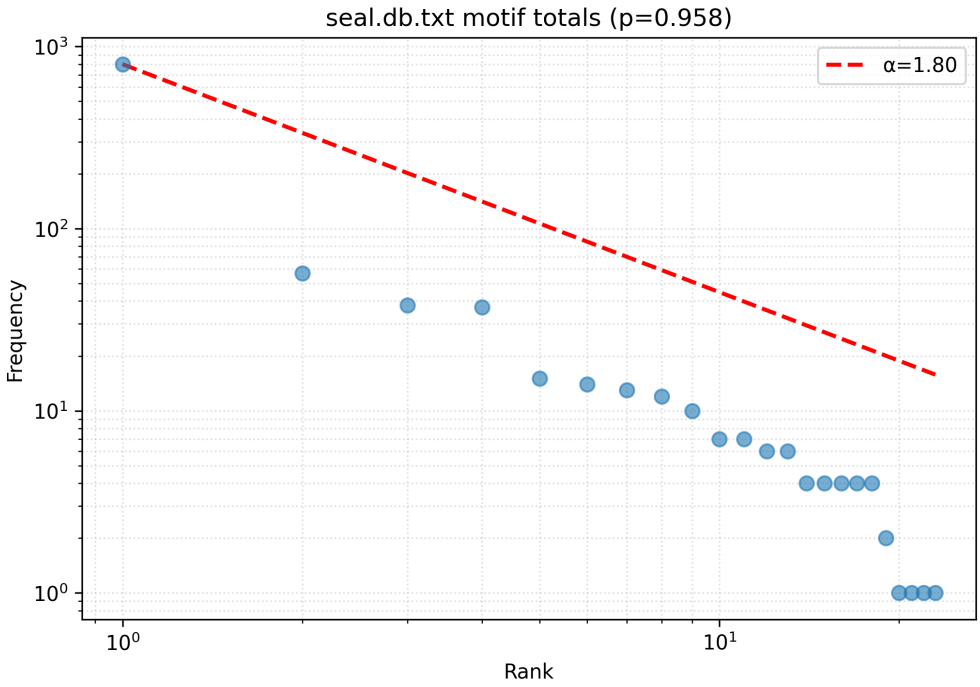


Figure 1: Log-log rank-frequency plot of the 23 iconographic motif classes (total seals per motif) with the fitted power law ($\alpha = 1.80$). The Unicorn motif dominates the head of the distribution.

The fitted exponent $\alpha \approx 1.80$ ($1.5 < \alpha < 2.0$) lies in the heavy-tailed, scale-free range. The power law is a plausible fit ($p = 0.958$) and decisively beats the exponential (Table 2), which the exponential's own bootstrap also rejects ($p < 0.0004$). It cannot be distinguished from the lognormal or truncated power law, as expected for so small a sample.

In the Bianconi–Barabási fitness model Bianconi & Barabási (2001), a node whose intrinsic fitness far exceeds its competitors captures a macroscopic share of all links. The Unicorn's share of over 76% of seals, across geographically disparate sites, is consistent with a single, highly fit standard outcompeting regional alternatives.

Several caveats apply. Seal counts are subject to recovery and preservation bias. Finally, small-type distributions in material culture can arise from neutral drift Bentley et al. (2004), but a single type holding a 76% share against 22 competitors deviates strongly from neutral expectation, pointing to selection-driven, winner-take-all dynamics.

5 Unicorn Seal System: Statistical Analysis

5.1 Data

We analyze published typological data from a sample of 500 unicorn seals across 19 archaeological sites, drawn from the comprehensive dissertation by Jamison (2017). In a comprehensive diachronic study, Jamison (2017) catalogued extensive stylistic variation in unicorn seals, decomposing the iconography into eleven attributes—offering stand, ear, halter, neck,

eye, head, hoof, tail, pizzle, leg, and horn—each with multiple recorded types.

Table 3 reproduces Jamison's numerical ranking of attributes by type count.

Table 3: Numerical ranking of stylistic attributes by count of types (after Jamison 2017, Table 7.3).

Attribute	Rank	Number of Types
Offering Stand	1	114
Ear	2	51
Halter	3	49
Neck	4	45
Eye	5	42
Head	6	39
Hoof	7	36
Tail	8	24
Pizzle	9	19
Leg	10	18
Horn	11	5

Jamison's typological work is invaluable; it focuses on the organization of seal production and its role in sociopolitical organization and stylistic variations in them. We argue that Table 3 itself reveals a functionally significant distinction. The offering stand has *more than twice* as many types (114) as the next most variable attribute (Ear, 51). The remaining attributes —all anatomical parts of the unicorn animal—cluster in a narrower range (5–51 types). This asymmetry is the key observation: a seal reader at a warehouse does not inspect the ear, eye, or hoof independently of the animal. One sees a unicorn. The anatomical variation reflects how different artisans rendered the same animal, not distinct encoded information. The offering stand, by contrast, is spatially separate from the unicorn and functionally distinct—it is the one non-animal, non-script element on the seal, and it is precisely the element with maximal variation.

We propose that this asymmetry reflects a functionally meaningful distinction in the seal's information architecture. Examination of the frequency distribution of offering stand styles reveals the signature of a power law with a characteristic heavy tail: a small number of dominant stand entity types appearing on many seals, and a long tail of rare types.

5.2 Attribute Distributions

We base our statistical analysis on the complete attribute data from Jamison (2017)'s Appendix 7.1 rather than the binned summary tables presented in his dissertation text. This distinction is methodologically important: Jamison (2017) reports 114 offering stand types in his dissertation narrative, whereas Appendix 7.1 records 139 offering stand types. This discrepancy likely reflects a later aggregation step in the narrative summary. For reproducibility, we treat Appendix 7.1 as the canonical dataset because it preserves seal-level attribute observations, allowing direct verification of type classifications and distribution fitting without

potential information loss from binning. The broader published corpus contains 500 seals, but not every seal contributes equally to every downstream test: some records lack usable stand or attribute values, and correlation analyses therefore work with smaller effective n values after pairwise deletion of 'n/a' and 'blank' entries.

Table 4 presents power law analysis results for all 11 seal attributes from Appendix 7.1.

Table 4: Power law parameters for all 11 seal attributes. Attributes are ranked by exponent α (ascending). All goodness-of-fit tests performed using 2500 bootstrap iterations. n_{types} = number of unique type classifications; n_{seals} = count of seals with valid data for that attribute; α = power law exponent; $x_{\min}$ = optimal lower bound; D = Kolmogorov-Smirnov distance; p-value = bootstrap goodness-of-fit p-value.

Attribute	Types	Seals	α	$x_{\min}$	D	p-value	GoF
Leg	18	449	1.4175	1.0	0.0714	0.8808	Yes
Tail	25	351	1.6333	2.0	0.1125	0.5708	Yes
Horn	5	443	1.6407	9.0	0.1623	0.9028	Yes
Halter	55	470	1.7537	2.0	0.0801	0.7452	Yes
Pizzle	20	436	1.7606	5.0	0.1287	0.5200	Yes
Head	39	431	1.8454	4.0	0.1016	0.5040	Yes
Stand	139	413	2.3591	3.0	0.0514	0.9764	Yes
Ear	52	438	2.4210	7.0	0.1035	0.4064	Yes
Neck	72	458	2.5234	7.0	0.0925	0.4700	Yes
Eye	44	417	2.9340	13.0	0.1589	0.0372	No
Hoof	38	442	2.9396	13.0	0.0655	0.8936	Yes

Methodological note: Horn exhibits an estimated $x_{\min}$ of 9.0 despite having only 5 total types, indicating that the power law fitting algorithm may have encountered instability with extremely sparse type data.

The unicorn's anatomical attributes all show heavy-tailed frequency distributions consistent with power-law or lognormal form, but their type counts are too few to separate the two. This is a limit of the data, not a finding of uniformity.

The critical observation is not distributional form per se but the magnitude asymmetry. The offering stand, spatially separate from the unicorn on the seal surface and not anatomically constrained by the logic of depicting a bovine, generates 139 distinct types against a unicorn-attribute baseline clustering in the 5–72 range. Even granting that anatomical attributes face physical constraints on variation the stand's excess type diversity requires a generative explanation beyond craft drift.

5.2.1 Two Generative Mechanisms Operating on Seal Attributes

The power law exponents and goodness-of-fit statistics reveal a fundamental distinction in how different seal attributes are generated, with important implications for understanding the information structure of the seal system.

Neutral Cultural Drift (BHS mechanism, $\alpha \approx 1.5$–2.0, Bentley et al. (2004)): Five attributes exhibit exponents characteristic of cultural drift: Leg ($\alpha = 1.42$), Tail

($\alpha = 1.63$), Halter ($\alpha = 1.75$), Pizzle ($\alpha = 1.76$), and Head ($\alpha = 1.85$). These attributes show the pattern predicted by Bentley et al. (2004) for neutral cultural transmission: each generation of craftsmen departs slightly from their training exemplar, producing heavy-tailed distributions in which a small number of conventions dominate and a long tail of rare variants accumulates. We note that the horn is diagnostic of a Unicorn and it drifts to its canonical form.

Preferential Attachment (BA mechanism, $\alpha > 2$, Barabási & Albert (1999)): Three anatomical attributes exhibit exponents above the BHS range, consistent with preferential attachment dynamics: Ear ($\alpha = 2.421$), Neck ($\alpha = 2.523$), and Hoof ($\alpha = 2.940$). All three pass goodness-of-fit tests (all $p > 0.40$). Attribute correlation analysis in the following section explores whether these high exponents reflect intentional bundling with stand identity.

Attributes Excluded from Analysis: Eye ($\alpha = 2.93$, $p = 0.0372$) failed the goodness-of-fit test ($\mathrm{p} < 0.05$), indicating its distribution does not follow a power law. Horn was flagged earlier for methodological instability (5 types with estimated $x_{\min} = 9.0$). Both are excluded from further interpretation.

5.2.2 Offering Stand Distribution

The offering stand is the most salient seal element that is spatially distinct from the unicorn motif, and it dominates all other attributes in type diversity (139 types) and in goodness-of-fit ($p = 0.9764$, the highest of the 11 attributes). Its exponent, $\alpha = 2.36$, lies in the preferential-attachment range (Barabási & Albert, 1999). Where nodes carry heterogeneous fitness, as commercial guilds do in reputation, commodity specialization and regional reach, the Bianconi–Barabási model (Bianconi & Barabási, 2001) predicts α in the range 2–3, consistent with the exponents observed for identity-encoding attributes ($\alpha = 2.36$–2.52). Hubs of differing fitness attract other nodes preferentially, so a small number of dominant guild types account for most seals, and a long tail of rarer identities reflects smaller or more specialised actors who link to the dominant hubs.

The bootstrap goodness-of-fit test passes the standard acceptance threshold ($p > 0.1$) of Clauset et al. (2009), supporting power law as a plausible generative model for the per-type distribution. The same test rejects the exponential ($p_{\mathrm{GoF}} < 0.001$, $0/2500$). The lower bound $x_{\min} = 3.0$ leaves $n_{\mathrm{tail}} = 46$ types in the power law regime, so $\alpha = 2.3591$ is a preliminary estimate; the evidence rests on the goodness-of-fit together with the linear log-log trend in Figure 2.

Table 5 summarizes likelihood ratio tests against four alternative distributions. No alternative rules out the power law (minimum $p = 0.108$, truncated power law), and it is favored over the exponential ($R = 1.25$, $p = 0.66$), though less decisively than for binned data because of the smaller tail sample.

Table 5: Distribution tests for offering stand data using complete Appendix 7.1 dataset. First row shows exponential goodness-of-fit bootstrap test (rules out exponential). Remaining rows show log-likelihood ratio tests: positive R favors power law; negative R favors alternative.

Test	R	p	Interpretation
Exponential (GoF bootstrap)	—	< 0.001	**Exponential rejected**
Lognormal (vs PL)	-0.90	0.340	No significant difference
Exponential (vs PL)	1.25	0.657	Power law favored, not significant
Stretched exponential (vs PL)	-1.07	0.319	No significant difference
Truncated power law (vs PL)	-1.29	0.108	No significant difference

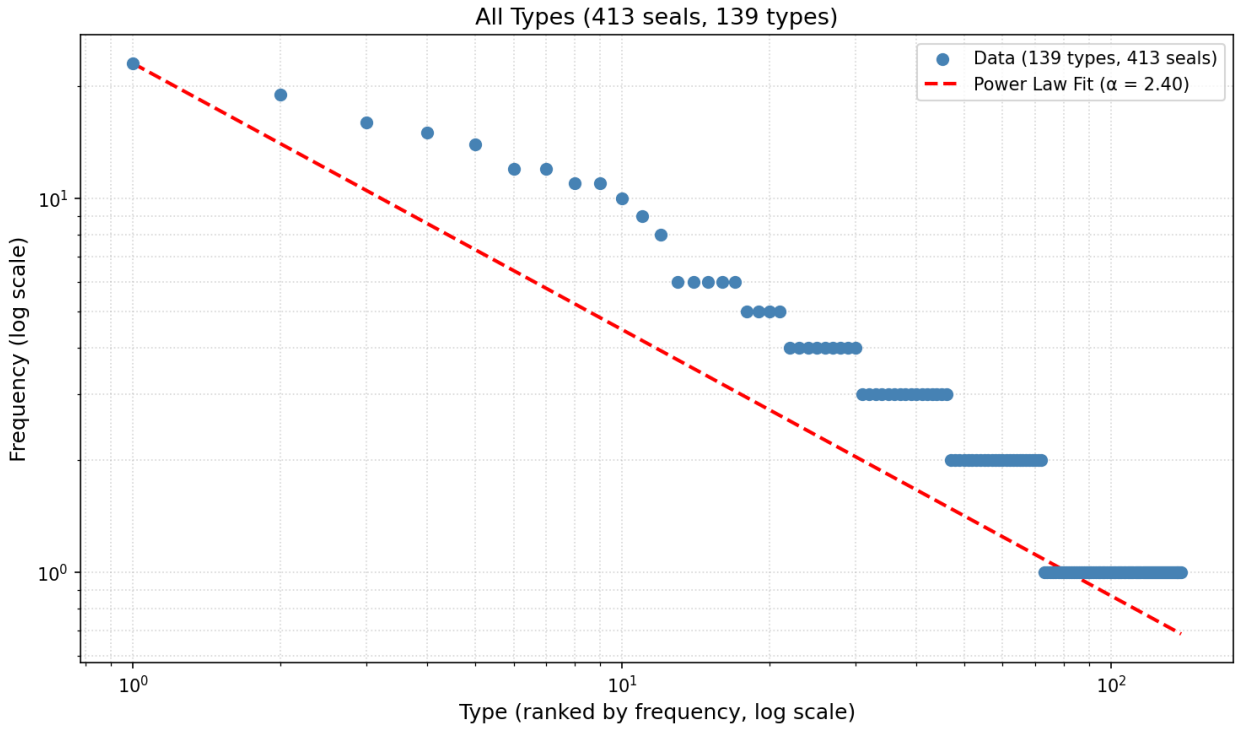


Figure 2: Log-log rank-frequency plot of all 139 offering stand types from the complete dataset (Jamison, 2017). The linear trend in log-log space is consistent with power-law structure despite the smaller tail sample ($n_{\text{tail}} = 46$).

5.3 Global Stand-Attribute Correlations

We ask two questions of the Appendix 7.1 attribute data. First, does stand identity predict the other Unicorn attributes across all usable seals? Second, does that association hold across sites, or could it simply reflect local carving habits? For the first question we compute the association between stand type and attribute value using Cramér's V (bias-corrected

Bergsma-Wicher, 0–1 scale; near 0 means independence). Permutation p-values with Bonferroni correction ($\alpha = 0.005$) are used throughout. Because each test excludes only seals with missing values for the relevant attribute, the effective sample size varies across rows.

Table 6 summarizes results for all 10 anatomical attributes, with valid n ranging from 379 to 407. Eight of ten attributes show significant stand-dependent variation ($V = 0.30$–0.48), which is strong for hand-carved, partly worn archaeological material. Leg and Halter show the strongest associations ($V \approx 0.43$–0.48). Eye ($V = 0.22$, $p = 0.066$) and Horn ($V = 0.29$, $p = 0.054$) do not reach corrected significance. Horn has only five types and is nearly constant across seals, so it carries little stand information; it is better read as a diagnostic feature of the Unicorn itself than as part of a stand-specific bundle.

Table 6: Global stand-attribute correlations using pairwise deletion. V = Cramér's V (bias-corrected Bergsma-Wicher); p_{perm} = permutation test p-value (5000 iterations); $\alpha = 0.005$ with 10 attributes (Bonferroni correction). Sample size varies by row because seals with 'n/a' or 'blank' entries are excluded only for the affected attribute. Attributes ranked by V (strongest associations first). Status: ✓ = significant at corrected α; ⚠ = weak (one test only).

Attribute	n	V	p_{perm}	Status
LEG	402	0.484	< 0.001	✓
HALTER	407	0.431	< 0.001	✓
PIZZLE	383	0.365	0.001	✓
TAIL	309	0.340	0.0038	✓
HOOF	393	0.337	0.0004	✓
HEAD	388	0.330	0.0008	✓
EAR	386	0.307	0.0002	✓
NECK	401	0.303	0.0008	✓
HORN	389	0.294	0.0538	⚠
EYE	379	0.221	0.0656	⚠

5.3.1 Cross-Site Bundle Agreement

A pooled association can arise from local carving tradition alone: if carvers at one site favor certain stand–attribute combinations and carvers at another site favor different ones, stand and attribute will correlate in the pooled data without any shared convention. The decisive test is whether the *same stand* carries the *same attribute value* at *different sites*.

For each attribute we take all pairs of seals that carry the same stand but come from different sites, and record the fraction of pairs sharing the same attribute value (the cross-site match rate). To judge this against chance, stand labels are randomly shuffled among seals within each site. The shuffle preserves each site's stand mix and attribute habits, so only a stand-linked pattern can exceed it. The procedure uses a fixed random seed for reproducibility. Results appear in Table 7.

Table 7: Cross-site bundle agreement. Match rate = fraction of same-stand, different-site seal pairs sharing the same attribute value. Null = mean match rate when stand labels are shuffled within sites. Different-stand = match rate for different-stand, different-site pairs. p is the permutation p-value (smallest attainable value ≈ 0.0005); Bonferroni $\alpha = 0.005$. Attributes with few types (Horn) match often by chance.

Attribute	n	Pairs	Same stand	Different stand	Null	p
HALTER	407	550	0.287	0.127	0.135	< 0.001
PIZZLE	383	514	0.280	0.123	0.127	< 0.001
HOOF	393	537	0.240	0.065	0.073	< 0.001
TAIL	309	347	0.239	0.159	0.156	0.0025
NECK	401	554	0.191	0.064	0.070	< 0.001
LEG	402	550	0.513	0.325	0.346	< 0.001
HEAD	388	538	0.152	0.076	0.080	< 0.001
EAR	386	540	0.139	0.030	0.034	< 0.001
EYE	379	473	0.116	0.048	0.049	< 0.001
HORN	389	535	0.761	0.715	0.721	0.136

Nine of ten attributes show cross-site agreement beyond the within-site null, all surviving Bonferroni correction. For Halter, the same stand shares a halter type in 28.7% of cross-site pairs against 13.5% expected from local habit; for Neck the figures are 19.1% against 7.0%, and for Hoof 24.0% against 7.3%. Horn is the exception, consistent with its near-constant form. The same stand therefore tends to carry the same attribute values at distant sites at two to three times the chance rate, which local carving tradition alone does not explain.

Three limits apply. Match rates are well below 1, so this is a shared convention with variation, not a fixed bundle per stand. Agreement is uneven across stands: among widely attested stands some match closely on an attribute while others, such as Stand #44, show almost none, and per-stand samples (roughly 8–22 seals) are too small for individual stand claims. Finally, a mobile, trained workshop tradition would also produce this pattern, so the result establishes a shared convention but does not by itself identify a coordinating institution; that inference rests on the trade context and the model developed later.

5.3.2 Tier-Specific Attribute Hierarchy

We next analyze attribute-stand associations across three practical commercial tiers: the top 5, 10, and 15 offering stand types. At each tier we compute Cramér's V between each attribute and the stand types present at that tier, asking whether the association persists as the market widens. Here again, n varies by attribute because the tiered analyses use only seals assigned to the relevant stand tier and then exclude 'n/a' or 'blank' values pairwise for each anatomical field.

Methodological note: The top-tier stands are predominantly multi-site operations. Stands #16, #32, #44, #1, and #30 each appear across 3–5 archaeological sites spanning 1000+ kilometers, and pooled V alone cannot separate a shared convention from local workshop tradition (see the cross-site test above). Because both n and the number of stand types

change from tier to tier, V values are not strictly comparable across tiers, and the tier trends are descriptive. If V *decreases* with tier size, core attributes stay coupled to stand identity while attribute diversity grows with market expansion.

Table 8 summarizes Cramér's V for all 10 attributes across the three tiers. All associations are significant at Bonferroni-corrected $\alpha = 0.05/10 = 0.005$.

Table 8: Attribute-stand association strength (Cramér's V) across three market tiers. n = number of seals; k = number of stand types present at each tier. All associations significant at Bonferroni-corrected $\alpha = 0.005$. Attributes ranked by mean V across all three tiers (strongest associations first). Decreasing V across tiers suggests core attributes remain tightly coupled despite increasing stand-type diversity.

Attribute	Top 5		Top 10		Top 15	
	$n = 86$	V	$n = 140$	V	$n = 171$	V
NECK	86	0.524	140	0.432	171	0.354
HALTER	86	0.492	139	0.389	171	0.395
HORN	85	0.332	138	0.371	168	0.464
HOOF	82	0.470	135	0.397	165	0.296
EAR	85	0.418	137	0.348	167	0.321
EYE	82	0.385	131	0.378	160	0.286
LEG	86	0.407	140	0.337	171	0.290
PIZZLE	84	0.397	136	0.302	165	0.223
HEAD	85	0.332	137	0.338	168	0.251
TAIL	65	0.374	108	0.290	135	0.218

The tiered pattern is descriptive. In the top five tier (stands with 23, 19, 16, 15 and 14 seals, together 87 seals or about 21% of the corpus), Neck ($V = 0.524$) and Halter ($V = 0.492$) are the most strongly stand-linked attributes, with Hoof third ($V = 0.470$). Neck then weakens as the tier widens (0.354 in the top 15), and Hoof falls more sharply (0.296), whereas Halter stays comparatively stable (0.395). Globally Leg and Halter lead ($V = 0.484$ and 0.431) and Neck ranks eighth ($V = 0.303$), so Neck's association with stand is concentrated in the hub tier. One reading is that the hub-tier bundle did not extend uniformly to rarer stands while Halter linked stand identity across the market; the tier comparison cannot establish this, since n and the number of stands differ by tier.

The pooled V results and the cross-site test (Table 7) together indicate a shared convention for pairing stand and attributes: eight of ten attributes associate with stand identity, and nine of ten agree across sites beyond the within-site null. The agreement is partial, so we describe a convention with variation, not a fixed bundle per stand. Whether a coordinating institution enforced it is an inference from the trade context and the model developed below, not a result of these statistics alone. We agree with Jamison (2017), who argues that "unicorn seal production was decentralized, it is important to reiterate what is actually meant by this statement. It does not imply that unicorn seal production and use were not regulated and controlled."

5.4 Diachronic Statistics

To examine whether the power-law structure is stable across the Harappan chronology, we assigned each seal in Appendix 7.1 a period using Tables 9.3, 9.5 and 9.6 of Jamison (2017) (merged file `appendix7.1.diachronic.txt`). Most seals were recovered before modern stratigraphic controls, so Jamison (2017) assigned periods by comparing seal and inscription carving styles with stratigraphically controlled HARP material from Harappa, the most reliable dating currently available for the corpus. The bootstrap goodness-of-fit test uses 1000 iterations with random seed 42. For period 3A (3 stand types, 4 seals), automatic $x_{\min}$ selection in `powerlaw` is unstable, so a fixed $x_{\min} = 1$ is used.

Table 9: Diachronic power law analysis of offering stand distributions by period. D = Kolmogorov-Smirnov distance; p = bootstrap goodness-of-fit p-value (1000 iterations, seed = 42). *Period 3A has only 4 seals and is not informative.

Period	Stands	Seals	α	D	$x_{\min}$	p-value
3A (2600–2450 BCE)*	3	4	2.6898	0.1163	1.0	0.5250
3B (2450–2200 BCE)	27	45	2.2649	0.1333	1.0	0.0160
3C (2200–1900 BCE)	42	93	2.0906	0.0665	1.0	0.2140

Estimated exponents exceed 2 in all three periods, but goodness-of-fit is uneven (Table 9, Figure 3). Period 3C is compatible with a power law ($p = 0.214$), period 3B fails the bootstrap test ($p = 0.016$), and 3A is uninformative. The diachronic samples are small, so these results are suggestive and additional stratified seals would allow a sharper test of when power-law structure emerged. Stand diversity rises from 3 to 27 to 42 types across the periods, which is consistent with expanding trade networks and growing commercial specialization, and the heavy-tailed form in the later phase suggests the stand system was not a late or ephemeral feature.

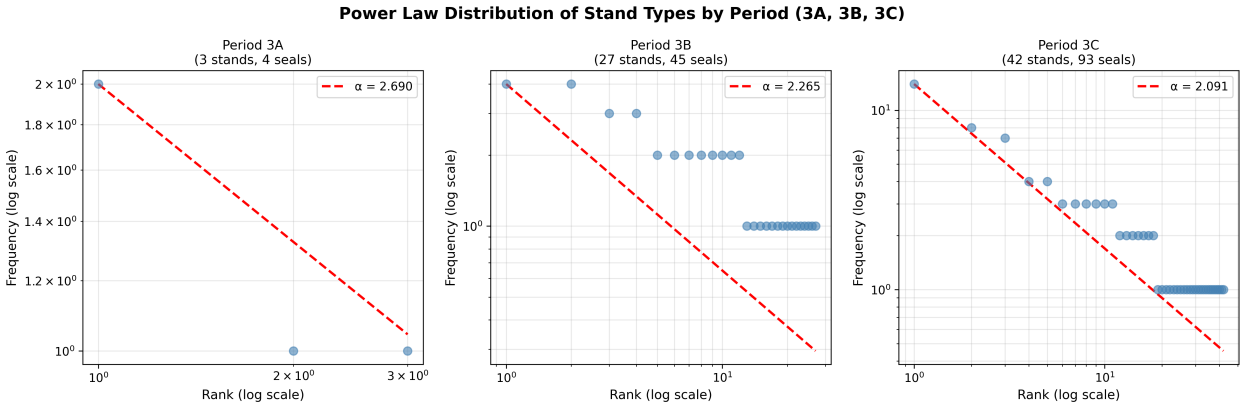


Figure 3: Diachronic rank-frequency distributions of offering stands across the three Harappan periods. The later phase is the most compatible with a heavy-tailed distribution; the earlier phases are limited by sparse data or weaker goodness-of-fit.

5.4.1 Diachronic Trajectory of the Unicorn Seal System (USS)

Seal counts and stand variety across the three periods are tabulated below.

	3A	**3B**	**3C**
USS seal count	4	45 (+1025%)	93 (+107%)
Stand variety (types)	3	27 (+800%)	42 (+56%)

The trajectory has the shape of an S-curve: a lag phase in 3A, rapid expansion in 3B, and slower growth in 3C. This is consistent with network-adoption dynamics in which new participants joined a standard whose value rose with its use (preferential attachment), and with a slowing of new stand types in 3C as the roster of commercial identities was largely covered, while seal production continued to grow. We treat this as a suggestive heuristic and not a fitted model. Three points cannot support a logistic regression, the counts come from a sample of dated seals and not from the whole system, and a larger 3C sample can by itself produce more stand types. More dated seals are needed to tell a robust growth curve from a sparse-sample artifact.

6 Site Distribution: Statistical Analysis

6.1 Data

Table 10 shows the distribution of analyzed unicorn seals across the 19 sites in the sample.

Table 10: Distribution of unicorn seals in the sample by site and state of preservation (reproduced from Jamison 2017, Table 7.1).

Site	Seals analyzed	Number of seals published	Number of intact seals
Harappa	193	312	110
Mohenjo-daro	191	832	408
Lothal	35	45	19
Kalibangan	24	26	13
Chanhu-daro	16	21	11
Bagasra	7	7	5
Dholavira	6	6	6
Allahdino	5	5	4
Balakot	4	4	1
Nausharo	4	4	4
Rakhigarhi	3	3	2
Banawali	3	3	2
Farmana	2	2	2
Nindowari	2	2	2
Jhukar	1	1	1
Kot Diji	1	1	0
Lohumjo-daro	1	1	0
Pabumath	1	1	0
Surkotada	1	1	1
Totals	**500**	**1277**	**592**

6.2 Site Seal Distribution

6.2.1 Geographic Distribution of Seals

The distribution of seal counts across archaeological sites follows a power law:

- Power law exponent: $\alpha \approx 1.55$
- Lower bound: $x_{\min} = 2$ (5 single-seal sites excluded)
- KS distance: $D = 0.094$
- Goodness-of-fit $p = 0.76$ (bootstrap, 2500 iterations)

The fit is strong ($D = 0.094$, $p_{\text{GoF}} = 0.76$) and outperforms an exponential distribution ($R = 9.09$, $p = 0.019$). The high goodness-of-fit p-value means the power law cannot be rejected as a plausible generative model for the data (Clauset et al., 2009). The optimal lower bound is $x_{\min} = 2$; the power law holds across fourteen sites with 425 seals (five single-seal sites excluded).

Table 11 summarizes the likelihood ratio tests against four alternative distributions.

Table 11: Distribution comparison tests for site seal counts. Positive R favors power law; negative R favors the alternative.

Alternative	R	p	Interpretation
Lognormal	-0.35	0.520	No significant difference
Exponential	9.09	0.019	Power law significantly better
Stretched exponential	-0.39	0.501	No significant difference
Truncated power law	-0.61	0.271	No significant difference

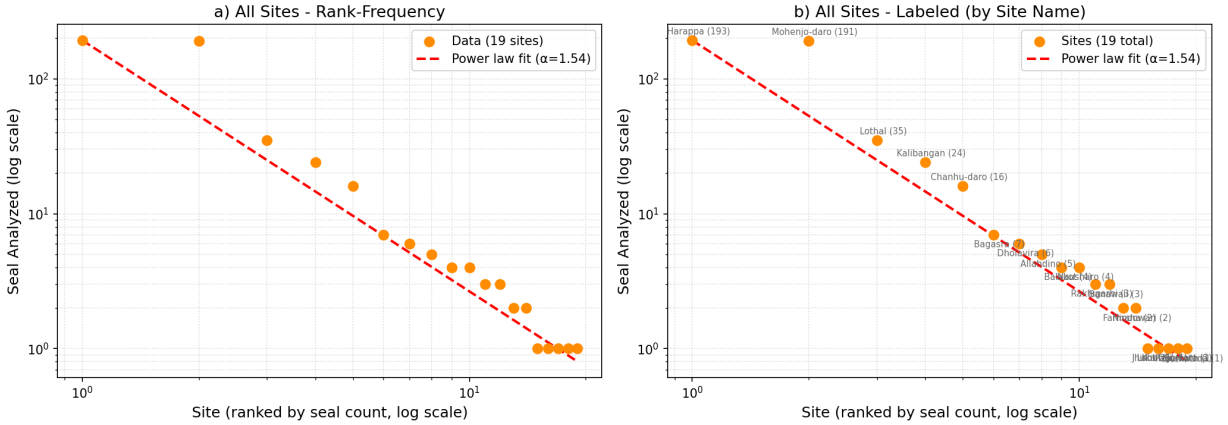


Figure 4: Log-log rank-frequency plot of seal counts by archaeological site. (a) All 19 sites. (b) Sites labeled with names and seal counts. Harappa and Mohenjo-daro emerge as clear hub sites.

The two most extensively excavated sites—Harappa (193 seals) and Mohenjo-daro (191 seals)—together account for 76.8% of analyzed seals from only 10.5% of sites. This concentration may partly reflect excavation history; sites such as Rakhigarhi, potentially the largest Harappan settlement by area, could alter the distribution as new seal data are published. The lower exponent ($\alpha \approx 1.55$) compared to the commercial entity distribution reflects a heavier tail and even greater concentration at the site level: the top few cities are disproportionately dominant trade hubs.

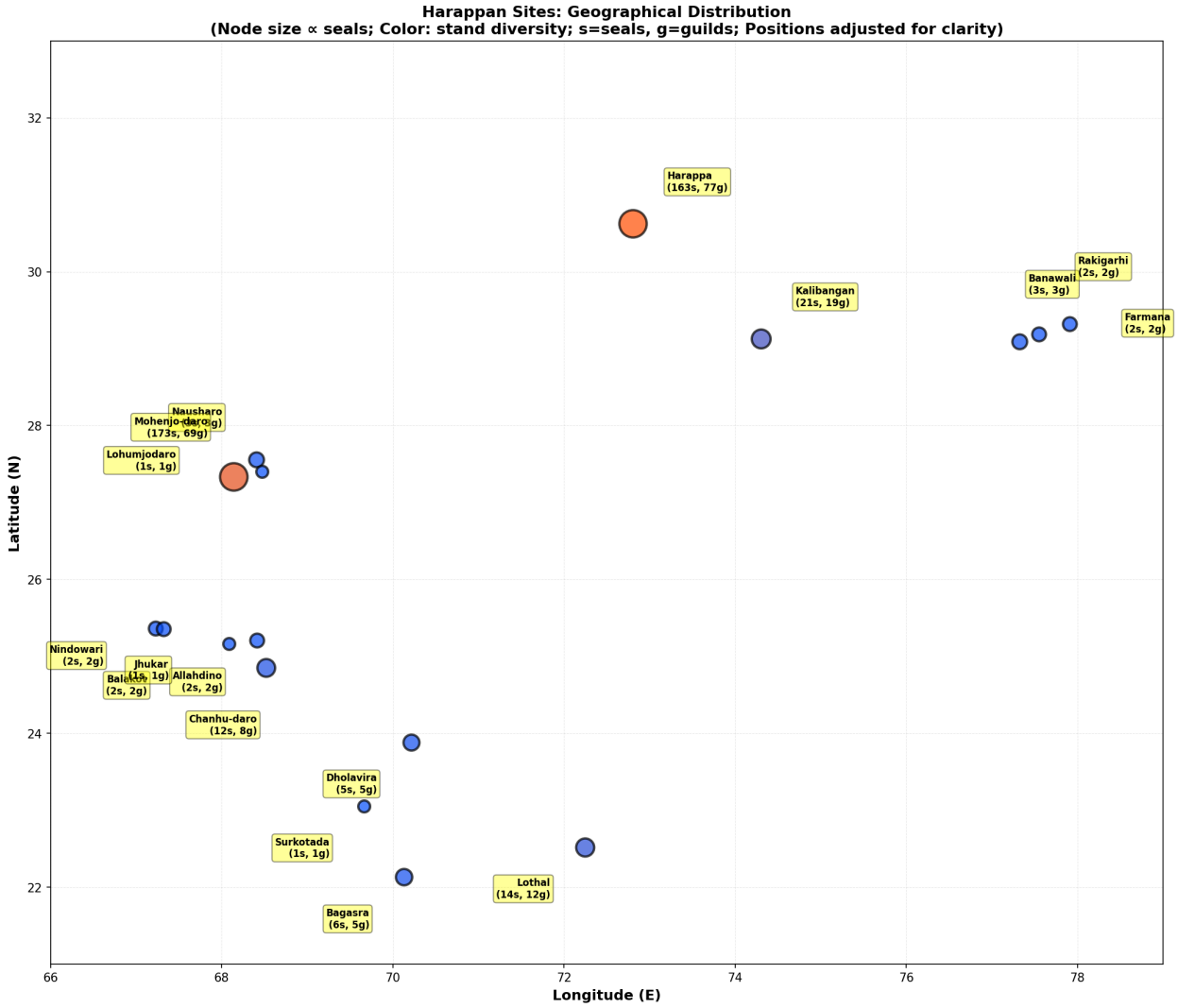


Figure 5: Geographic distribution of seal types across archaeological sites.

6.2.2 Multi-Site Hub Guilds and Network Topology

Of the 139 stand types, 64% are recorded at one site and 36% at two or more. Stands #25 and #32 each occur at five geographically dispersed sites, while Harappa and Mohenjo-daro share the greatest overlap in stand types (Figure 6). These distributions indicate unequal geographic reach among stand types, although they cannot by themselves establish branch offices or direct trading relationships.

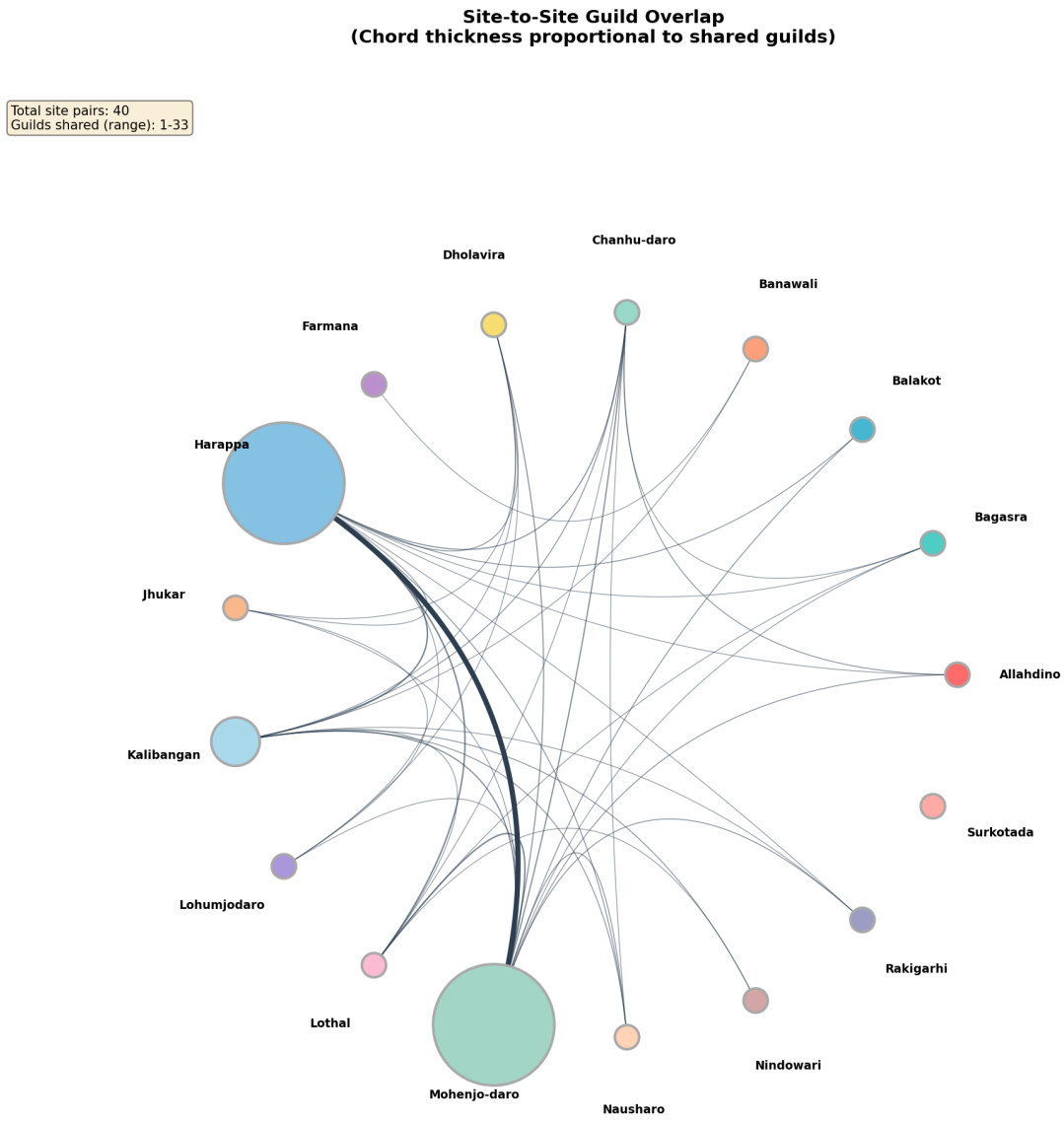


Figure 6: Offering-stand overlap between archaeological sites. Chord thickness represents the number of shared stand types; node size represents each site's seal count.

6.2.3 Singleton Guild Distribution

The geographic distribution of singleton stands—stand types appearing at only a single site—reveals a complementary pattern that illuminates the structure of Harappan commerce. Figure 7 maps the spatial distribution of these 50+ unique stand types, showing that singleton stands are not concentrated at a single hub but instead are distributed across the

entire geographic network, including peripheral and satellite sites.

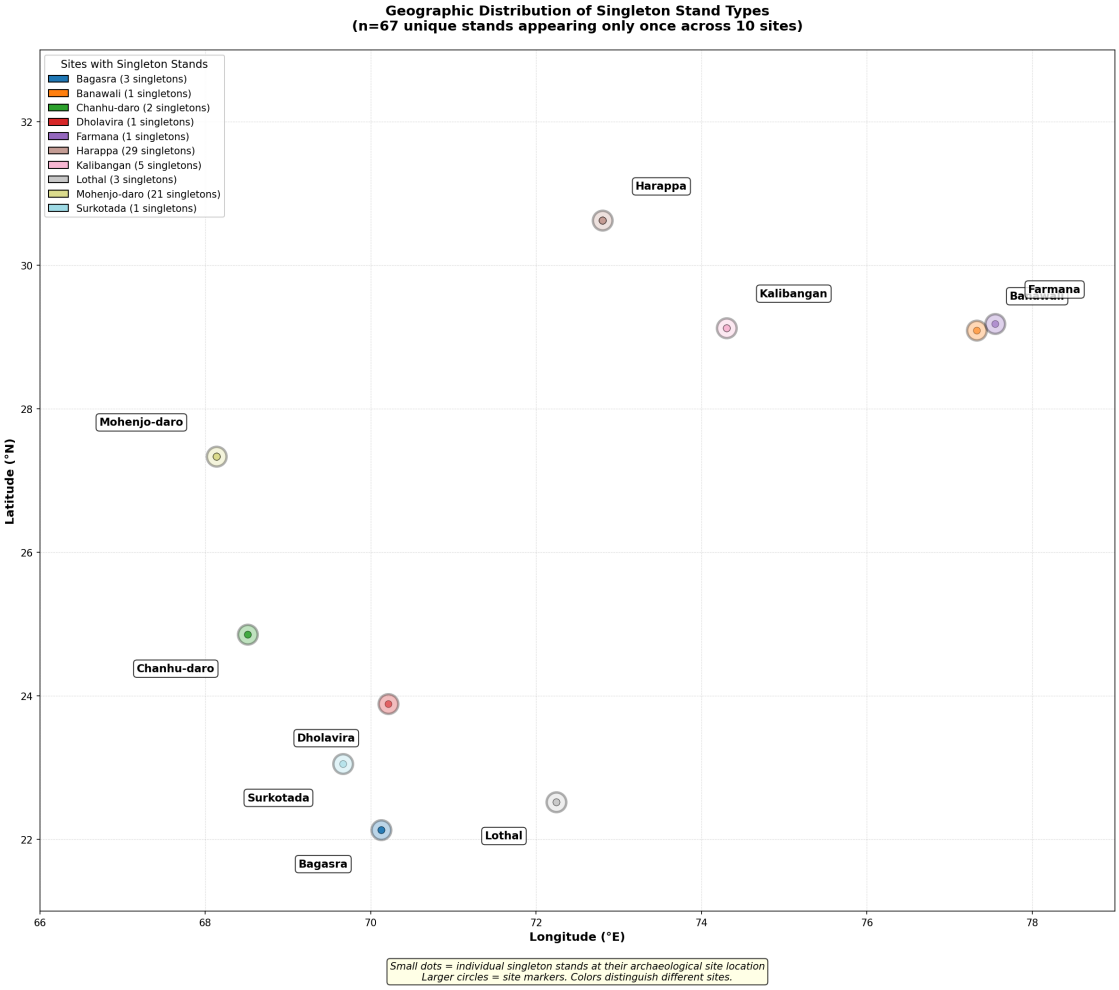


Figure 7: Geographic distribution of singleton stands across Harappan sites. Singletons (stand types appearing at only one site) are widely dispersed across the network rather than concentrated at hub locations.

7 Archaeological grounding of the statistical analysis

The seal system of the Indus civilization evolved over nearly a thousand years as increasing trade generated problems related to scaling in long-distance commerce. This evolution was not just decorative drift—it was iterative engineering that extended the reach and reliability of trade networks across unprecedented distances.

7.1 Seal Evolution as Trade Scaling Solution

Archaeological work by Kenoyer & Meadow (2010) and colleagues, based on excavations at Harappa and other sites, documents three distinct phases of seal development. We argue that these phases correspond to changing trade network demands.

7.1.1 Geometric Seals (Early Harappan, c. 3200–2600 BCE): Local Trade among Known Parties

The earliest Indus seals bore geometric designs—intersecting circles, cross-hatching, concentric squares, swastika and similar motifs. These seals appear in context with clay sealings at sites such as Harappa (Kot Diji phase), demonstrating use in verification and authentication from the earliest pre-urban phases. Geometric seals perhaps functioned effectively for small, local networks where parties knew each other personally and relationships could possibly be family-based. In these networks, agreements were verbal and did not require written documentation. The seal indicated ownership and distinguished goods from similar consignments, and the sealing (clay impression with cord marks) provided tamper-proof evidence of package integrity during short-distance movement as well as record-keeping.

Material evidence reflects this early phase: geometric seals were manufactured from diverse materials—bone, terracotta, steatite, and faience—suggesting experimentation. This trial-and-error approach gradually converged on steatite, which offered a crucial advantage: when fired at high temperature, steatite becomes hard and glassy, allowing seals to be used for longer term, perhaps persisting across repeated trade cycles. This firing innovation provided the first long-term contract with perhaps an anti-counterfeiting mechanism. Steatite seals are indicative of centralization and standardization in trade practices, reflecting the growing complexity of economic interactions.

Additionally, seals and sealings were small and portable—ideal for record keeping in preliterate communities and easily carried. At this stage, perhaps the number of sealings at the receiving end could serve as a rudimentary record of incoming goods between counterparties. Parties could meet seasonally to reconcile records by simply matching the number of sealings each party held and settle accounts.

7.1.2 Animal and Plant Motifs (Mature Harappan, c. 2600–2200 BCE): Expansion Beyond Family Networks

As trade scaled beyond local, kinship-based networks, the geometric system reached a critical bottleneck: geometric patterns could not reliably distinguish between different traders or trading houses. Shippers, transport agents, and their workers would have had difficulty determining where packages should be routed. The system required clearer identity markers.

The solution was the adoption of real totemic animal and plant motifs—bull, elephant, rhinoceros, tiger, gharial, and various plant designs. These motifs carried cultural resonance within their respective ethnic or regional communities, functioning as recognizable emblems for specific trading houses and families. Each could be assigned to a guild or merchant collective, serving as a brand identifier and indexing product lines or specialization.

However, this system too faced a critical scaling problem: the number of real animals, plants, and their recognizable variants is finite. More importantly, as the trade network expanded to draw merchants from diverse cultural backgrounds, the use of totemic animals could create political issues. Additionally, coordinated seal production was needed to prevent issues where different parties could choose the same iconography, and to establish something neutral to avoid trade frictions related to routing and handling.

7.1.3 The Invented Unicorn (c. 2600–1900 BCE): Breaking the Cardinality Ceiling

The unicorn—a fantastic creature with bull-like body, single forward horn, slender proportions, and no natural prototype—appears suddenly in the Mature Era and dominates approximately 60% of seals thereafter. This motif was likely an administrative or commercial invention, born from the convergence of two pressures: the exhaustion of real animal variants and the need for a politically neutral symbol. Unlike real fauna and flora that carried totemic significance to particular cultural groups, the unicorn belonged to none. It functioned as a compromise animal—a creature that could represent network membership without according primacy to any single cultural or totemic group.

The invented unicorn and the stand broke the cardinality ceiling: the repertoire became extensible beyond the finite inventory of recognizable real animals and plants. More importantly, the unicorn signified membership in the confederation itself, analogous to a modern corporate logo or franchise mark. Its universality conveyed that a seal-user belonged to the confederation united across the Indus valley—not because of ethnic, regional, or totemic affiliation, but through voluntary commercial participation and confederal membership. Each merchant guild retained its distinct offering stand as a marker of individual identity and guild autonomy. The exceptional diversity of offering stands in the corpus—139 distinct types across 19 sites—provides direct evidence of the confederation's scale and inclusive membership structure: each stand variant represents a merchant guild or trading collective admitted to the confederation, making the stand repertoire itself a registry of the confederation's expansive, pluralistic architecture where autonomy and unified standardization coexist.

7.2 Archaeological Evidence for the Trade

The seal evolution documented by Kenoyer & Meadow (2010) and colleagues is grounded in stratigraphic and contextual evidence. But to understand how this system actually operated—where seals were deployed, how they were managed, and what operational patterns they reveal—we turn to detailed archaeological documentation of seal distribution and use contexts. Based on this evidence, we can propose a plausible operational architecture of the Indus commercial network.

7.2.1 No Anthroponyms and Toponyms

Mukhopadhyay (2023) establishes that Indus script inscriptions encode neither anthroponyms nor toponyms. Our statistical findings confirm this conclusion. This is crucial: it means the seal script system could not identify individuals or places by name. Instead, identity had to be encoded through non-linguistic means: the unicorn motif (commercial network marker), the offering stand variant (guild affiliation), and the accompanying attribute bundle (confirmation checksum).

7.2.2 Seal Distribution and Association

Archaeological evidence confirms that seals served trade related logistic functions. Mukhopadhyay (2023) documents that inscribed seals concentrate in functionally specific contexts: near city gates, craft workshops, and public buildings, often alongside standardized Indus weights. Sealings appear attached to storage containers and warehouse locking systems. These contexts are uniformly commercial and logistic—gateways for traffic control, workshops for production control, warehouses for access control. The seals functioned to enforce "rules involving taxation, trade/craft control, commodity control and access control" (Mukhopadhyay, 2023). They were not decorative, not ritual, not administrative records (which would require centralized archiving). They were tokens deployed at points of commercial decision-making.

7.2.3 Trade Control Architecture

Frenez & Tosi (2005) provides the most detailed archaeological documentation of sealing practices from Lothal. His work documents 93 individual sealings recovered from a warehouse context, with 70 concentrated in a single square meter (Block C, SRG3, Layers 2–3), representing a logistical checkpoint. He notes a pattern: across the Indus civilization, the seal-to-sealing ratio is approximately 10:1 (over 3,700 seals known, but only 200 tablets and 130 sealings total). Frenez notes that this ratio contrasts sharply with Mesopotamian evidence, where sealings are nearly as abundant as seals and are maintained in administrative archives. The Lothal cache itself was preserved accidentally by fire—the only reason such a sealing archive survives in the archaeological record. This scarcity of sealings raises a fundamental operational question: why would seals be preserved and reused, while sealings appear only rarely and must be reconstructed from accidental preservation?

Fuls (2023) cataloged Mound F at Harappa and documents a large inventory: 173 square seals, 36 rectangular seals, 164 bas-relief tablets, and 215 inscribed tablets. Kenoyer and Meadow interpret this as warehouse storage. Crucially, the Mound F cache contains no sealings—only the permanent tokens (seals and tablets). This pattern aligns with Frenez's findings: seals and tablets persist in the archaeological record, while sealings are ephemeral. Together, Frenez and Fuls provide evidence consistent with a trade control architecture.

8 Core Inferences from Statistics and Archaeology

The statistical patterns of the Unicorn Seal System (USS) are consistent with a self-organizing network rather than a centrally controlled administrative system. We take this as a point of departure for understanding the Harappan civilization as a trade-driven commercial network spanning a vast region. The statistics alone do not establish this reading. It rests on the trade contexts in which seals and sealings are found, and on the absence of rulers or religious imagery in the archaeological record, which makes a commercial interpretation the most parsimonious one.

In the scale-free network models of Barabási & Albert (1999) and Bianconi & Barabási (2001) discussed earlier, new entrants preferentially link to established hubs rather than to peers at random. This produces a self-reinforcing dynamic: successful merchants attract

more partners, which increases their reach and reputation (fitness) and attracts further partnerships. Merchants seeking to minimize risk and maximize access to reliable partners would naturally converge on established hubs. Aggregated over hundreds of merchants and many generations, such decentralized choices can produce a scale-free structure, with a small number of hub guilds linking many peripheral traders. Such a topology requires no central authority; it can emerge from individual commercial choices and network effects (Katz & Shapiro, 1985; Arthur, 1989).

Further clarification, in network science terms, when seals are interpreted as nodes representing commercial entities or trading houses, a direct link or shared interaction implies an active trade relationship within a Barabási-Albert (BA) or Bianconi–Barabási (BB) network topology.

8.1 Claim #1: Offering Stand as Commercial Entity Emblem

We argue that each unique offering stand type (139) is the emblem of a persistent group engaged in trade, an identifier that maintained relationships across sites and time within the USS. The key fact that it is visually seperate on the seal face, is the most diverse non-animal motif and it follows power-law distribution that conforms to a node in a trade network. Over the roughly 700 years of the Mature Harappan period, the same stand types recur across sites, which points to persistent group identities rather than one-off artisanal choices. We read these groups as commercial entities because stands occur in trade contexts and because no competing religious or royal interpretation is visible in the record. Other group-level interpretations, such as kin or clan marks, could produce similar distributions, so this reading rests on the archaeological context rather than on the statistics.

8.2 Claim #2: Unicorn Seal System as a Winning Standard or Technology

The predominance of the USS, together with its power-law form, suggests a seal-based standared or technology that came to prevail in the Mature phase of the Harappan civilization, in accordance with network-externality theory. The diachronic data are consistent with increasing participation by commercial entities within the USS (4, 45, and 93 seals across the three periods). Arthur (1989) describe the process: "when two or more increasing-returns technologies 'compete' for a 'market' of potential adopters, insignificant events may by chance give one of them an initial advantage in adoptions. This technology may then improve more than the others, so it may appeal to a wider proportion of potential adopters. It may therefore become further adopted and further improved." The USS is such a standard that was adopted by more and more of the commercial entities described in the previous claim.

8.3 Claim #3: Commercial Entity Participation in Trade Contracts or Agreements

As described above, commercial entities participated in trade links connecting distant sites across the Harappan trade network. Each of these commercial entity participated in form-

ing a link with another entity with top few of them becoming the hubs of the network in accordance with network theory. We propose that a pair of such seals, one for each party, would serve as the record of an agreement or contract between them in the network. We infer this from network science as well as trade related context where seals are found. We cannot identify specific trading partners archaeologically, but the network science is consistent with the such a preferential linkage. The wide distribution of some top stand types across sites indicates the breadth of that entity's trading participation. Some stands appear at many dispersed sites, indicating extensive trading links; others appear at a single location, indicating limited reach or peripheral status.

8.4 Claim #4: Coordinating Institutional Requirements

The cross-site agreement between stands and attributes (Table 7) indicates a shared convention for pairing stand with animal features, maintained across hub cities and distant settlements. A convention of this kind suggests some form of coordination. The agreement is partial rather than uniform, so the evidence supports a shared standard with variation, not strict enforcement. Hub guilds may have developed the USS cooperatively as a trade and logistics innovation (Katz & Shapiro, 1985), since they had the resources to manufacture seals, administer the system, and enable smaller traders to participate. Singleton stands, found at only one site, plausibly lacked such resources, yet they carry the same convention; how they obtained properly made and validated seals is the coordination question. A trained, mobile workshop tradition could also produce this pattern, but given the trade context a coordinating institution is the more parsimonious explanation. This is the another hypothesis of the paper, and it is an inference from the statistics together with the archaeology, not a result of the statistics alone.

Such an institution or federation would need to:

- validate new entrants seeking seals, prevent counterfeiting, and keep each stand unique so that it identifies its holder durably;
- provide seal infrastructure to independent producers, through manufacturing, logistics coordination, or licensing;
- ensure protocol compliance by enforcing standards across participants;
- resolve disputes arising from contractual arrangements.

The evidence does not specify whether these functions were concentrated in a single organization, distributed among guilds, or exercised through a commercial federation. The most frequent stands (#16, #1, #30, #32, #44) may have had disproportionate influence in setting standards, but the long tail of smaller participants indicates that the system was not a closed group. We hypothesize that the Unicorn Logistics Confederation (ULC), described in the model developed below, is such an institution, and that its role helps explain the Unicorn's dominance in seal iconography. The following section examines whether the archaeological record preserves the operational traces such a system should leave.

8.5 Claim #5: Scale-Free Signature and self organization

A heavy-tailed, power-law-like signature appears at several levels of Harappan glyptic art: in the overall seal motif (Unicorn and others) distribution, within the Unicorn Seal System itself with stand types, and in site-level counts. This is suggestive of self-organization as an organizing principle in the Harappan civilization that recurs at many levels of sealing technology.

9 Seals, Tablets and Sealings: Asynchronous Barter System and Clearing Exchange

The Harappan trade network spanned more than a million square kilometers, the key challenge is one of logistics of a variety of packages type to be safely conveyed multi-modally as well as via multiple parties spanning diverse regions of the Indus area. We propose that the Harappan sealing system played a central role in coordinating this complex logistics as well as accounting within the trade network.

The statistical patterns and archaeological contexts above motivate a hypothetical model organized around three coordination challenges:

- **Routing across distance and language:** Long-distance exchange could involve multiple transport modes and linguistic communities. We propose that the USS's two-layer information system combined rapid visual identification and routing with more detailed, specialist-readable information.
- **Reconciliation over time:** Because shipments were asynchronous, merchants needed to review which consignments had arrived, which remained outstanding, and how obligations balanced. Recurring meetings on an agreed schedule could provide a time and place for reconciliation, renewed agreements, and dispute resolution.
- **Secure handover between parties:** A proposed protocol had to preserve identity and accountability as goods passed from Party A through coordinating logistics providers to Party B, and in reciprocal exchange from B to A. We model the ULC as coordinating that process, without assuming the archaeological record proves such an institution existed.

These challenges organize the hypotheses developed below: the USS's two-layer information architecture, periodic meetings, the Asynchronous Peer-to-Peer Barter System (AP2PBS), and the Unicorn Logistics Confederation (ULC). The archaeological record does not directly document these institutions or practices.

9.1 Two-Layer Information Architecture

We propose that the statistical correlation within the USS is consistent with it being designed in two distinct layers:

Table 12: The seal's two-layer information architecture.

	Layer 1 — Iconography (Stand)	**Layer 2 — Script**
Audience	Package sorters, dockworkers, gatekeepers, handlers of goods	Guild members, scribes, trade specialists
Content	Guild emblem (stand) + visual confirmation checksum (neck/halter etc.)	Product detail, quantities, types, measures
Function	Rapid visual verification of guild identity and authenticity	Precise record-keeping and contractual terms
Skill required	None—readable without literacy across regions	High—specialist-readable, needed only at origin/destination

This two-layer design solves a fundamental Bronze Age logistics problem: authenticating goods across vast geography without literacy while enabling reliable package routing and claiming without visible centralized administration. Layer 1 is universal and fast and hard to forge; Layer 2 is secure but requires specialists. This dual-layer system specifically was iteratively arrived at to solve the multi-modal logistic challenge of long distance barter.

9.1.1 Attribute Correlations: A Visual Confirmation Protocol

Packages were secured with clay impressed by the seal. Intermediaries at way-stations, caravan stops and ports had to validate package identity from sealings for onward routing, across linguistic and cultural zones. When a sealing was damaged or illegible, handlers could fall back on visual pattern-matching rather than script; a package would also carry multiple impressions for redundancy. We propose that the offering stand served as the *primary visual identifier*, quickly distinguishing one merchant's packages from another's, and that the characteristic attributes (neck, halter, ear, etc.) paired with it formed a *confirmation protocol*. A handler would check the attributes against the expected bundle for that stand—"Does this Stand 16 display the expected Halter variants? Yes, so route to Vessel A"—allowing rapid sorting where intermediaries could not read script. A constrained bundle also deters fraud, since an imposter would struggle to reproduce all of it. At origin and destination, parties presumably knew the script and could verify seals through several channels; the visual bundle was redundant there but reinforced identity. The stand-attribute correlations and the cross-site agreement reported earlier are consistent with such a convention, but do not show that it was used this way.

9.2 Periodic Meetings: Reconciliation, Renewal, and Innovation

In this trade driven model, periodic meetings brought merchants together to manage the parts of trade that could not be completed through asynchronous shipment alone. At recurring gatherings, participants could reconcile outstanding obligations, and resolve disagreements over delayed, damaged, or disputed consignments. By netting reciprocal claims, they could close one cycle of exchange.

These gatherings would also provide occasions to negotiate new agreements and revise existing terms. Merchants could respond to changing supplies and demand by setting new prices or exchange ratios, agreeing on quantities and quality standards, and introducing newly available products to trading partners. Repeated face-to-face contact could renew relationships and trust, while shared decisions could maintain compatible practices across the network. In this sense, meetings were not only settlement events but also forums for commercial coordination and innovation.

The archaeological record does not identify this protocol or institution, so their periodicity and functions remain inferential.

9.3 The Unicorn Logistics Confederation (ULC): Institutional Functions

The ULC's value proposition rested on several crucial functions that individual merchants could not provide independently without mutual coordination:

- **Guaranteed weight standardization:** The ULC maintained and distributed uniform weights (hexahedral chert stones) across all major sites. This ensured that when Party A contracted to send "100 standard units of carnelian," Party B could verify the quantity using identical weights at their location. Without standardized weights, transactions would fail due to disputes over quantity.

- **Controlled seal manufacture and authentication:** The ULC controlled seal production and authorized them. A logistic entity encountering an unfamiliar seal could trust that if the ULC had authorized it, it belonged to a recognized guild. The production of seals may have been decentralized but still following ULC standards for stand and attribute bundle as well as repertoire of script.

- **Dispute settlement and arbitration:** When discrepancies arose—missing goods, damaged shipments, unmet contractual terms—the ULC provided neutral arbitration at seasonal settlement conferences. Parties brought their evidence (tablets with discrepancy markings, sealings), and ULC arbiters negotiated resolutions binding on all parties.

- **Secure delivery guarantee:** The ULC physically transported goods, provided waypoint security, and vouched for chain-of-custody. If goods were lost or stolen in transit, the ULC bore liability (subject to proof of delivery via sealings).

- **Counterparty risk underwriting:** This was the crucial function. Before the ULC, a merchant sending goods faced absolute risk: what if the counterparty never sent their promised goods back? The ULC solved this by vouching for both parties' good faith. By placing its reputation behind every transaction, the ULC reduced counterparty risk to manageable levels. Without this underwriting, stranger-based trade over such vast distances would have been economically irrational.

- **Multi-modal logistics coordination:** The ULC maintained expertise across diverse transport modes—carts, pack animals, river boats, maritime vessels—and knew optimal routing through different seasons and ecological zones. Individual merchants need not master all this knowledge; it would be impractical to do so individually. The ULC provided this expertise.

These functions explain why any guild would participate in the ULC—these services are inferential and not directly observable. The form of compensation remains uncertain: participation may have involved fees, licensing payments, membership contributions, reciprocal obligations, or other benefits exchanged for access to the ULC's services. Whatever the precise arrangement, the ULC solved coordination problems that isolated merchants could not solve alone, making participation valuable even for guilds that could not maintain independent long-distance infrastructure. Archaeological evidence from Kotada Bhadli (Kachchh) identifies possible caravanserai infrastructure, suggesting that such way-stations and logistics nodes existed to support long-distance trade networks and lending support to the hypothesis that a coordinating institution like the ULC would have maintained physical sites for goods staging, waypoint verification, and merchant assembly (Shirvalkar et al., 2025).

9.4 Operational Model: Asynchronous Peer-2-Peer Barter System (AP2PBS)

We propose an operational model that could explain this evidence: the Asynchronous Peer-to-Peer Barter Settlement (AP2PBS) protocol. AP2PBS is a merchant-developed protocol that emerged from the collaborative efforts of multiple guilds seeking to coordinate trade across unprecedented distances and cultural boundaries. The proposed AP2PBS protocol explains how seals, tablets, and sealings were used to actually handle the operational lifecycle of long-distance barter and settlement. AP2PBS provided more than a sequence for moving goods: it anchored trust across an almost million-square-kilometre geography. AP2PBS made distant exchange operationally legible; the ULC made participation in that system credible.

The following five phases outline how merchants, aided by the ULC, conducted asynchronous peer-to-peer barter across vast distances.

9.4.1 Phase 1: Conference—Initial Agreement

Merchants A and B meet face-to-face (at a conference venue, market hub, or port) to negotiate a future barter transaction:

- **Negotiation of terms:** Merchants agree on goods, quantities, and quality standards. Example: "A will send 100 carnelian beads of grade X to B; B will send 50 grain sacks of quality Y to A."
- **Agreement inscription on seals:** The agreed terms are carved into unique seals. Each merchant designs or commissions a seal bearing their guild's unicorn + offering stand + confirmation attributes, with script encoding the product and quantity metadata.

- **Tablet exchange:** Each merchant produces two clay tablets impressed with their own seal (which have their guild's identifying attributes) and gives both tablets to the other party. These tablets are "claim vouchers" as well as "routing manifest". The "claim voucher" B is for A to claim goods from B delivered by the ULC. The "routing manifest" B is for A to send their goods with sealing A which the ULC will route to B. This is the simplest form of a peer-to-peer barter arrangement.

  They could be multiple such pairs of tablets each indicating mutual obligations that each party could keep as a record of their obligations across multiple cycles. We note here that these tablets may have perforations, allowing them to be strung and attached to the packages for usage as a routing manifest.

- **Parties depart:** Merchants return to their home settlements. The tablets are their portable proof of obligation. Without these tablets, the ULC will not release goods at the destination.

9.4.2 Phase 2: Shipment—Origin Verification and Packing

Party A produces the agreed goods and prepares them for transit:

- **ULC witnessing and verification:** ULC agents examine the goods, verify their quality and quantity against standardized weights, and confirm the match with the agreed terms carved on A's seal.

- **Package sealing:** ULC agents wrap the package with cordage and seal it with Sealing A—an unfired clay lump impressed with A's seal. The sealing proves that the goods match the agreement and provides tamper-proof evidence of package integrity.

- **Tablet handover:** Party A hands Tablet B (bearing Party B's seal impression) to the ULC agents. This tablet becomes the ULC's routing document—it specifies the final destination and recipient. It likely was strung to the package for routing purposes. It would have to be strung to the package throughout its journey separate from the sealing itself to maintain the integrity of both the routing information and the package seal.

- **Departure:** The package (bearing Sealing A) and Tablet B (held by ULC agents) embark on transit. They will travel together to Party B's location but remain physically separate.

9.4.3 Phase 3: Network Routing and Intermodal Transit—ULC Management

In Phase 3, the package (bearing Sealing A) and Tablet B (the routing manifest) traverse the network from origin to destination under ULC management. This phase is the operational heart of AP2PBS—where the stand + confirmation bundle, statistically demonstrated throughout our analysis as the core of the seal system, becomes a functional logistics tool.

- **Stand + Confirmation bundle as transactional identity:** The seal on Tablet B –the unicorn plus offering stand plus attributes (Cramér's V >0.45 correlation)– constitutes the confirmation bundle. This bundle is the primary identifier across the entire network. Each guild maintains a distinctive stand composition and attribute set, visually recognizable at any waypoint across all sites and languages. This confirmation bundle is not decorative; it is logistics infrastructure coordinated by the ULC. It allows ULC handlers at multilingual waypoints to authenticate the shipment and forward it without reading script.

- **Permanent ULC site agents as network nodes:** At each major site (Harappa, Mohenjo-daro, Dholavira, Lothal, Shortugai, etc.), the ULC maintains a permanently stationed site agent. This agent is not a visiting merchant but a professional logistics operator—multilingual, culturally versed in local practices, and embedded in the local community. The site agent's responsibilities form the operational backbone of the system: receiving shipments, verifying seals and tablets, holding goods in secure storage (warehouse), releasing consignments to outbound carriers or to the final recipient, managing weight standards, witnessing discrepancies, and gathering market intelligence on local trade flows and merchant behavior.

- **Market intelligence and counterparty monitoring:** The permanent site agent is the network's eyes and ears. By observing which merchants are sending goods to which destinations, in what quantities and commodities, the agent gathers intelligence on market patterns, seasonal flows, and merchant relationships. This information flows back to senior ULC coordinators at hub sites. If a merchant suddenly fails to send expected goods, if patterns shift dramatically, or if disputes arise, the site agent's observations provide evidence for ULC arbitration (Phase 5). This function also serves an early-warning purpose: if a guild begins to show signs of financial distress or unreliability, the site agent can flag it, allowing the ULC to reduce counterparty risk exposure.

Link to ULC Functions: Phase 3 operationalizes all ULC functions from Section 7.2. The permanent site agents enforce weight standardization (**Function 1**), authenticate seals via the confirmation bundle (**Function 2**), document discrepancies as future arbitration evidence (**Function 3**), manage multimodal transport routing (**Function 6**), and gather the intelligence needed to underwrite counterparty risk (**Function 5**).

9.4.4 Phase 4: Two-Key Delivery Handover—Mutual Authentication

Party B (or their agent) appears to claim the goods. Dual authentication ensures both parties are legitimate:

- **Party B's tablet check:** Party B presents Tablet A as well as their own seal to ULC agents. ULC agent matches the presented Tablet A with Sealing A and confirms Party B's seal against the routing authority token (Tablet B) that Party A handed to the ULC.

- **Package sealing comparison:** ULC agents compare the seal impression on Tablet A (presented by Party B) with the impression on Sealing A (the package seal). These must match. If they match, Party B is legitimate and authorized to receive the goods.
- **Mutual authentication:** Simultaneously, ULC agents present Tablet B to Party B. Party B examines this tablet and confirms it bears their own seal—proof that the ULC agent is the genuine, authorized carrier sent by the original agreement.
- **Re-weighing and discrepancy marking:** ULC agents open the package in front of Party B as an official witness. They re-weigh the contents against standard weights. If shortages or damage are found, Party B marks these discrepancies directly onto the back of Tablet A using a stylus (incising scratches to indicate units missing or damaged).
- **Sealing A as receipt:** ULC agents take Sealing A (only one, though packages may be stamped with multiple sealings bearing A's seal for redundancy), as proof that delivery occurred as contracted. Party B receives the goods and keeps Tablet A (with or without discrepancy markings). Tablet A is now a completed record of one leg of the transaction by Party A.

Similarly, Party B would complete asynchronously their leg of the transaction.

9.4.5 Phase 5: Reconciliation and Settlement—Periodic Clearing

At the turning of seasons, all principals reconvene at a central clearing venue:

- **Token assembly:** Merchants bring their physical evidence. Party A brings Tablet B (with or without discrepancy markings); Party B brings Tablet A (with or without discrepancy markings). ULC agents bring delivery sealings.
- **Obligation reconciliation:** Parties lay out tokens and reconcile. If Party A's Tablet B shows no discrepancies and Party B's Tablet A matches the goods received, the obligations cancel. If discrepancies exist (missing goods, damage), parties negotiate adjustments: partial refunds, compensatory goods in next cycle, or arbitration awards.
- **Deliberate destruction:** Once multi-party debts are squared and balanced to zero, the physical tokens (tablets and sealings) lose all legal value. To prevent fraud and double-counting, they are intentionally destroyed—broken, discarded, or thrown into architectural fills and refuse deposits at the settlement venue. At this time if agreements breakdown or there is need for newer agreements, they destroy old seals.
- **Network reset:** The transactional cycle closes. Accounts are cleared. The platform resets for the next seasonal wave of trade. Parties depart, ready to negotiate new agreements. Phase 1 of the next cycle then begins with the creation of new agreements and corresponding seals and tablets.

9.5 Archaeological Evidence: Reinterpreted according to AP2PBS

9.5.1 Seal Distribution and Guild Identities

Mukhopadhyay (2023) documents seal contexts across the Indus civilization systematically. Her observations, compared with the AP2PBS model, show convergence on core points and demonstrate where AP2PBS adds operational detail. She proposes administrative and taxation functions; we determine that commercial function is a more parsimonious explanation, given the lack of evidence for an extractive or centralized state in the archaeological record. The following table presents this comparison:

Table 13: Mukhopadhyay's archaeological observations vs. AP2PBS operational interpretation. Mukhopadhyay establishes WHAT seals, tablets, and sealings were found and WHERE. AP2PBS explains HOW they functioned in practice.

Artifact/Context	**Mukhopadhyay's Finding**	**AP2PBS Operational Role**
Seals (distribution)	Concentrated at gates, workshops, storage facilities with standardized weights	Permanent guild identity tools; ULC agents carry seals at each waypoint for authentication
Identical iconography or inscriptions on seals, tablets and sealings at distant sites	Standardized regulation, permits, licensing	Asynchronous logistics: Seals at origin; tablets at destination (matching pairs); AP2PBS phases;
Identical iconography with differing inscriptions on seals	Officials/artisans/ merchants of same guilds/rulers/governing-bodies controlled/practiced differing crafts/trade	We agree but frame it as commercial differentiation of products
Differing iconography with similar inscriptions on seals	Officials/artisans/ merchants of different guilds/rulers/governing-bodies controlled/practiced similar crafts/trade	We agree but frame it as commercial competition for same products
Inscribed sealing-pendants of Kanmer	Conjectured to be passports and gate-passes	Asynchronous logistics: Seals at origin; tablets and sealings at destination (matching pairs); AP2PBS phases; additionally, these may have served as high-security escorts for valuable consignments; or perforated tablets could have been used as the routing tablet
Similar tablets at distant sites	Bear seal impressions on obverse; smooth backs or with numerical-metrological notations. License and tax marking	Routing documents (Tablet B) held by ULC; claim vouchers (Tablet A/B) for counterparties; smooth backs allow discrepancy marking; see AP2PBS phases
Seal + weight association	Seals and weights co-occur at gates/workshops	ULC agents verify goods weight against standard measures before sealing (Phase 2) and after delivery (Phase 4)
Script (no names/places)	Script encodes neither anthroponyms nor toponyms	Script encodes product metadata only; guild identity via seal, not written name; routing via tablet, not written address
Iconography	Mukhopadhyay documents seal contexts systematically, proposing administrative and taxation functions	Permanent guild identity. We confirm this via our statistical data - with commercial entity interpretation
Disappearance after IVC decline	Seals/tablets/sealings cease ca. 1900 BCE	Trade system collapse: without ULC coordination, AP2PBS has no operational basis

9.5.2 Frenez's Warehouse Evidence: Logistics and Operational Patterns

Frenez & Tosi (2005) documents 93 sealings from Lothal's warehouse excavation context in detail. His observations of sealing types, container associations, and spatial clustering reveal patterns consistent with asynchronous logistics. We interpret Frenez's empirical data through the AP2PBS operational model to show how these warehouse patterns could have functioned in long-distance trade coordination. The following table maps Frenez's documented findings to AP2PBS phases:

Table 14: Frenez's Lothal sealing data interpreted through AP2PBS operational model. Frenez documents archaeological patterns; AP2PBS explains their functional meaning for trade logistics.

Sealing Pattern	**Frenez's Observation**	**AP2PBS Interpretation (Phases 1–5)**
Multi-seal sealings (up to 5 seals per sealing)	L-189, L-190 bear 4 seals each	Multiple guilds' packages staged together in warehouse pending consolidated shipment (Phase 3 waypoint staging)
Dense warehouse cluster (70 in one square meter)	Sealings concentrated in Block C, SRG3, Layers 2–3	Logistical queue: outbound shipments awaiting transport; temporary accumulation before dispatch (Phase 3)
Diverse containers (wooden boxes, pottery, sacks, canes)	L-161–172 (boxes), L-177–181 (pottery), L-126 (sacks), L-134–140 (canes)	Different commodity types require different containers; ULC managed multimodal packing (Phase 2–3)
Fingernail tallies on boxes	L-161–172 bear 1–6 tally marks	Discrepancy records: units missing, damaged, or reconciled at delivery (Phase 4 re-weighing)
Seal reuse (13 seals, 76 impressions)	High-frequency seals (L-6: 16 impressions; L-37: 12 impressions)	Durable guild seals reused across multiple shipments; contrasts with one-time-use sealings (Phase 2 creation, Phase 4 handover)
Pegs-on-wall sealings	L-179, L-206 with vegetal fiber ropes	Storage room locks for batched goods: temporary security during Phase 3 warehouse staging
Unfired sealings + accidental fire preservation	70 sealings preserved only by fire at Lothal	Sealings are disposable: routinely destroyed after use (Phase 5 clearing); fire preserved only rare intact examples
10:1 seal-to-sealing ratio (across Indus)	5,000 seals known vs. 130–140 sealings	Seals are permanent tools (reused across many cycles); sealings are temporary labels (created per-shipment, destroyed at settlement)

The Lothal sealings were also classified with respect to their find spots to be from the "warehouse" context. It is also noted that this area was near the place where boats docked - again reinforcing the trade context of these seal, sealing, and tablet finds. Frenez (2024)

classifies most of the finds to the "warehouse," where he identifies multi-stamped sealings - we interpret these as packages awaiting shipment before loading onto boats that would eventually carry them.

9.6 Predictions and Confirmations of AP2PBS

The AP2PBS model generates a series of testable predictions.

9.6.1 Stand and Script commonality

The seals bearing the same stand iconography could be tested to see if they have common script. It would suggest that the commercial entity specializes in a particular class of goods or trade activity, perhaps with multiple partners.

If a particular stand type had a large inventory of scripts, it would likely indicate the breadth of products or transaction that guild had.

9.6.2 Trade Linkages

Identical inscriptions or iconography on seals and tablets found at sites separated by hundreds of kilometers are consistent with two parties holding complementary routing and claim tablets at opposite ends of a trade arrangement. In the AP2PBS model, one party's tablet authorizes a claim or provides routing authority for the other party, while the corresponding seal or tablet authenticates the consignment at the other end. Such a match links the two sites by at least one leg of the trade, without by itself demonstrating a complete route, a single commercial owner, or the full sequence of intermediaries between them.

9.6.3 Venues for Periodic Settlement and Conference

AP2PBS barter arrangements require periodic settlement meetings in a large, non-residential structures capable of hosting multiple merchants for periodic conference and settlement would be expected. Functional venues could include spacious halls with aligned seating, and nearby storage areas for goods staging. Such venues would not be temples or palaces but explicitly commercial infrastructure.

Pillared hall, as found in Mohenjo-Daro, may have served this function, providing a covered space for merchants to reconcile tablets and sealings in addition to negotiate, and settle disputes.

9.6.4 Scarcity of Tablets and Sealings Relative to Seals

Because settled pairs are deliberately destroyed, tablets and sealings should be far less common than seals (which are permanent long term contracts). Frenez (2024) reports the ratio of 10:1 seals to sealings. Over 3,700 seals are known, but only approximately 200 tablets and a similar number of sealings—Lothal being the largest single deposit. This ratio is exactly what AP2PBS predicts: reusable seals create thousands of tokens; disposable tablets and sealings create only dozens of archaeological traces.

9.7 Why AP2PBS Explains the Unicorn Dominance

As velocity of trade increased, the network expanded. More merchants, more guilds, more products, more trade links. The unicorn's dominance is therefore a direct consequence of AP2PBS and ULC driving network expansion.

This functional trade related innovation is consistent with what the quantitative evidence documents. The diachronic data reveal a distinctive statistical signature: rapid adoption cascade followed by plateau, consistent with preferential-attachment dynamics in an expanding network. The network's observed growth trajectory is the expected outcome hen a decentralized coalition of independent traders adopts a politically neutral, scalable trade protocol.

10 A Functional Reinterpretation of Harappan Civic Architecture

10.1 Form Follows Function: Trade Infrastructure, Not Temples and Palaces

The monumental buildings of Mohenjo-daro have often been interpreted through priest-king paradigms derived from other Bronze Age civilizations. Earlier excavators and subsequent summaries variously described the Pillared Hall as an "assembly hall" or "prayer hall," the Great Bath as a temple or ritual bathing space, and the so-called "Great Granary" as a facility for administrative or grain storage. These labels, however, remain functional hypotheses rather than demonstrated identifications. More recent archaeological interpretations have questioned the older palace, temple, and granary models and have emphasized the uncertainty of the buildings' original functions.

The absence of securely identified palaces, royal tombs, monumental royal inscriptions, and clear priestly iconography makes a centralized priest-king interpretation difficult to sustain. At the same time, the Harappan record is rich in evidence for organized production and exchange: seals, standardized weights, tablets, sealings, warehouses, workshops, gateways, and extensive transport infrastructure. Read together, these materials support interpreting Mohenjo-daro's monumental structures within the wider commercial and logistical system rather than assigning them inherited political or ritual functions.

Applying Louis Sullivan's dictum—**form ever follows function**—we therefore interpret these structures as possible components of the operational infrastructure of AP2PBS, or of a comparable long-distance barter system. This interpretation remains a reconstruction rather than a direct archaeological identification, but it explains the buildings within the broader evidence for distributed commercial coordination.

10.2 Three Functional Venues

10.2.1 The Pillared Hall: Phase 5 Settlement Chamber

Physical evidence:

- 27.5 metre square; 20 brick pillars in 4×5 rows, spaced 9'7" ± 3 inches (extraordinary precision)
- Ordered rows of seating in parallel aisles between pillars
- South-to-north slope in the rows of aisles—it has been suggested that the area was washed
- Paved walkways; upright bricks along aisles (designed for water drainage)
- Flat, polished stone (61×54×37 cm, dark brown) with clay-pressing wear patterns

Architectural Logic—Why This Geometry? A conventional assembly hall, religious gathering space, or administrative venue requires only an open floor plan, courtyard, or tiered seating, but does not require 20 massive pillars arranged in a rigid 4×5 grid. Such a grid imposes geometric order on human movement and partitions space into parallel aisles—a design suited to repeated, simultaneous, independent operations rather than collective gathering. The pillars themselves are not merely structural necessity; they are logistics infrastructure. This geometry solves a specific operational problem: enabling multiple independent trading pairs to reconcile accounts in parallel, in orderly stations, without interfering with one another. A parliament or temple does not require this. AP2PBS Phase 5 reconciliation logic suggests that obligations were netted out in these parallel trading stations and tablets and sealings were systematically destroyed and later washed away.

Function (Phase 5 reconciliation):

- Two counterparties + ULC agents sit in ordered rows with Tablets A/B and sealings
- Match claim tablets with package sealings; document discrepancies
- Destroy settled pairs (crushed tablets, broken sealings)
- Wash debris away via drainage bricks
- Press new Phase 1 tablets on polished stone; exchange vouchers
- Each rows of aisles accommodated a many such trading pair—parallel operations at scale

The polished stone as evidence: The repeated wear pattern on the polished stone surface indicates sustained, regular pressing activity. Within the AP2PBS framework, this is consistent with decades of clay-tablet pressing during Phase 5 reconciliation cycles. Mackay's alternative (leather-working) remains possible on physical evidence alone, but the wear pattern sits less naturally with the hall's other diagnostic features.

10.2.2 The Great Hall: Market Intelligence Hub

Scale & location: 230 feet × 78 feet; massive open space with multiple rooms. Wheeler's granary hypothesis rejected by Kenoyer's re-excavation (found zero grain). Architecture rules out residential, temple, or storage use.

Function (seasonal conference):

- **New product showcase:** Merchants introduce goods; agree on new seal templates and product codes
- **Market intelligence exchange:** Prices, harvests, demand signals from distant hubs (Magan, Dilmun, regional nodes)
- **Coordination among partners:** Periodic gathering of trading partners, plausibly supporting relationship maintenance across cycles.
- **Inclusive participation:** Smaller guilds participate alongside hub merchants; share technologies and market intelligence

Though this function is inferential rather than directly evidenced.

The scale and layout of the Great Hall—ruled out for residential, temple, or storage use—are consistent with a venue for periodic large-scale commercial gathering.

10.2.3 The Great Bath: Essential Hygiene Infrastructure

Physical infrastructure: Watertight tank (12m × 7m × 2.4m deep); bitumen-sealed; colonnaded corridors; elevated, set apart. Industrial-scale, not luxury.

Operational rationale: Long-distance transit left arriving merchants, cart drivers, and warehouse handlers covered in salt, silt, and dust—an unavoidable consequence of maritime, riverine, and overland cartage. The need for bathing, however, did not depend solely on periodic settlement conferences. The daily operation of a major commercial center—receiving raw materials, processing goods, loading and unloading consignments, handling animals and carts, and dispatching outbound shipments—would itself have created a continuous occupational hygiene requirement. In the hot climate of the Bronze Age, without modern ventilation, cooling, or sanitation infrastructure, workers engaged in these activities would have needed regular access to water for washing and recovery from dust, salt, sweat, and industrial residue.

Before Phase 5 reconciliation, in which counterparties sat face-to-face for extended negotiation and physical exchange of tokens, bathing would also have served a social and presentational function. The Great Bath can therefore be understood as practical civic infrastructure supporting the daily logistics of the commercial center, with periodic meetings providing an additional reason for its use rather than its sole purpose. Within the AP2PBS reconstruction, it was not merely a luxury or ritual amenity but part of the hygiene infrastructure required by sustained Bronze Age production, procurement, and exchange.

Dual function (proposed):

- **Possible social function:** A shared facility of this scale may also have served to mark the transition from transit to commercial engagement, though this remains inferential
- **Hygiene:** Removal of salt, silt, and dust accumulated in transit, consistent with the scale and industrial construction of the tank

The Harappan emphasis on bathing infrastructure—individual house baths, extensive drainage, and this monumental public facility—is more parsimoniously explained as civic infrastructure supporting commerce at scale than as a religious or purely domestic amenity.

10.3 The Integrated Commercial Precinct: Seasonal Cycle

Three structures formed a single operational system:

Table 15: Mohenjo-daro commercial precinct: AP2PBS operational roles.

Venue	Role
Great Bath	Hygiene + (possible social function, inferential + trust renewal)
Great Hall	Market intelligence + periodic partner coordination
Pillared Hall	Phase 5 settlement: matching, netting, destroying, signing
Polished Stone	New contract pressing

Seasonal merchant/workmen journey:

1. Arrive filthy (cart/boat/maritime transit)
2. Bath: cleanse, mingle with returning merchants
3. College Building: rest and lodging (function proposed; Kenoyer notes this structure's original use remains uncertain, dropping the unevidenced "priestly" attribution assigned by early excavators)
4. Great Hall: exchange market intelligence, ideas, planning, discuss new products, etc.
5. Pillared Hall: settle past-cycle accounts, match tablets & sealings, destroy reconciled records
6. Polished stone: press new Phase 1 tablets, exchange vouchers
7. Bath: final cleansing before departure

Year after year, decade after decade. This cycle was the proposed operational backbone of a peer-to-peer network spanning a million square kilometres—with no evidence of kings, priests, or central bureaucracy.

10.4 Logistics Hubs: Waterfront Warehouses

Beyond the Mohenjo-daro precinct, AP2PBS required staging infrastructure at nodes throughout the network. The so-called "granary" or "warehouse" structures at Harappa, Lothal, and other hub sites show no evidence of agricultural storage; read functionally, they are consistent with inbound and outbound logistics warehouses— the physical instantiation of Phase 3 network routing. Their architecture, when read functionally, reveals a complete workflow for goods reception, verification, staging, and dispatch.

10.4.1 Harappa's Mound F: The Inbound and Outbound Logistics Warehouse

Once identified as a granary, the Mound F structure's function remains uncertain: excavations have found no evidence that grain or other foodstuffs were stored there. Marshall (1931) describes a large brick complex with sleeper walls and ventilated rooms, on the northern edge of the mound beside the dry bed of the Ravi River, linking it geographically to riverine transport and Harappa's hinterland. The assemblage includes approximately 173 square seals, 36 rectangular seals, 164 bas-relief tablets, and 215 inscribed tablets, but no comparable accumulation of sealings (Fuls, 2023). This combination is consistent with a facility where durable identity tools and tablets were handled or stored while temporary package sealings moved on or were removed. It supports a logistics interpretation, but does not establish the building's function or the proposed workflow.

10.4.2 Lothal: Warehouse as Sealing Archive

Frenez (2024) documents 93 sealings in Lothal's warehouse context, about 70 concentrated within one square metre in Block C, Layers 2–3; some bear multiple seal impressions. This concentration may represent packages temporarily staged together before handling or dispatch, and an accidental fire preserved the sealings. In contrast with Mound F's few sealings, Lothal may capture a moment when temporary package records had accumulated rather than been removed. The difference is consistent with distinct stages of activity, but does not by itself demonstrate a shared AP2PBS procedure.

10.4.3 Kotada Bhadli: Hinterland Caravanserai and Distributed Way-Stations

Shirvalkar et al. (2025) identifies features at Kotada Bhadli in Kachchh consistent with caravanserai infrastructure, while evidence of dairy production may indicate provisioning for people or animals. The site could have supported rest and movement along overland routes connecting larger centres, complementing riverine and maritime hubs such as Mound F and Lothal. Its role as a transfer point within the proposed network remains an interpretation, not a directly documented function.

11 Discussion

11.1 Network Effects and Emergent Standardization: From Commonwealth to Coordinated Scale

The paradox posed in the Introduction is how the Harappan Civilization achieved extraordinary standardization (USS, calibrated weights, gridiron cities, uniform brick, coordinated drainage) without centralized political authority. We propose that network effects operating on independent merchant enterprises resolve it.

The statistical evidence from Sections 5 and 6 converges on three observations. First, the Unicorn Seal System is a shared authentication convention across 19 sites and 139 offering stand types, each stand functioning as a merchant guild or trading entity. Second, the diachronic trajectory (Section 5.4.1) is consistent with an S-curve of sparse emergence,

rapid expansion and slower growth. Third, the power-law distribution of offering stands (Section 5.2.2) is consistent with preferential attachment, as new traders copy a successful commercial convention.

These patterns are what network externalities would produce. When independent merchants converge on shared protocols because adoption becomes individually rational, standardization emerges from the bottom up, with an S-curve adoption trajectory and a power-law frequency distribution. No central authority need have issued orders: merchants adopted the convention because it widened their trading opportunities and counterparty confidence, and once critical mass was reached each adopter reinforced the standard through positive feedback. The result is coordination without a coordinator, akin to railways, interstate highways, telecommunications and the internet, where standardization is likewise diagnostic of network externalities. The hubs of such a network, the multi-site stands and the large sites, can be read as the "heterarchies" that Vidale (2018) describes as "the power systems that ruled the polycentric" cities of the Harappan Civilization: several centers of influence without a single apex. The institutional requirements of this self-organization are detailed in the next section.

11.2 Seal System and associated AP2PBS produces scale-free signatures

All the distributions—seal motifs, guild sizes and site seal counts—exhibit the heavy-tailed, hub-dominated structure characteristic of scale-free networks. The evidence rests on multiple pillars:

1. **Seal motifs follow a power law:** the Unicorn dominates the iconographic repertoire and the other motif classes fall off rapidly; the exponential is rejected, while lognormal and truncated power law cannot be distinguished from the power law in this small sample.

2. **Guild sizes follow a power law:** a few dominant guilds account for a disproportionate share of seals, while many guilds appear only once.

3. **The power law persists across periods:** guild-size distributions remain heavy-tailed from 3A to 3C, so the hub-and-spoke structure is not an artifact of a single period, although the early samples are small.

4. **Stand and attributes form a shared convention:** stand type is strongly associated with the other attributes, and the same stand tends to carry the same attribute values across sites, which is consistent with a coordinated standard.

5. **Site seal counts follow a power law:** a small number of mega-hub sites dominate, although excavation history contributes to this.

6. **Unicorn adoption is consistent with an S-curve:** a lag, rapid expansion and slower growth, suggestive of network effects and standard diffusion.

7. **Top guilds and singletons both span sites:** top-ranked guilds appear across most sites, consistent with high connectivity, and singleton guilds also appear at multiple sites, which is possible where common infrastructure exists.

This power-law structure—market hierarchy encoded in offering stand variants —explains how asynchronous peer-to-peer trade could operate at scale. Hub guilds (few in number, high in volume) could coordinate across the network. Peripheral guilds (many in number, low in volume) could participate through hub intermediation. The scale-free architecture made AP2PBS operationally feasible: a merchant from a peripheral guild could trade with strangers anywhere in the network by leveraging hub guild authentication and logistics infrastructure.

11.3 The Unicorn Seal as Multi-Layer Operational Contract

The unicorn seal was not a decorative symbol or administrative marker. It was a **multi-layer operational contract** encoding the full commercial transaction lifecycle:

Table 16: Unicorn seal functions across AP2PBS workflow phases - both directions.

Form	**Function**	**AP2PBS Role**
Seal on steatite (permanent)	Long running contract & guild identity	Phase 1 negotiation; Phase 1 sealing; reusable across cycles
Sealing A/B (impressed on package)	Integrity proof; tamper-evident; executed delivery	Phase 2 and 4 shipment; proves origin, integrity and sealing receipt
Tablet A/B with flat back	Claim voucher + discrepancy record + routing slip	Phase 3 routing; Phase 4 claim voucher; Phase 5 reconciliation; bilateral marking of shortages

Each seal impression is both **unique** (specific transaction) and **replicable** (same seal carved by same carver maintains visual continuity). This dual property allowed:

- **Authentication across time:** A seal carved once could be reused for decades, allowing merchants separated by years to verify their counterparty's identity
- **Confirmation across distance:** The confirmation bundle (unicorn + offering stand + attributes) could be recognized at any waypoint without language intermediation
- **Legal continuity:** Jamison's multi-site seal groups (21 of 55 groups span multiple sites, 38% cross-site distribution) represent guild networks persisting across decades and geography

The offering stand variants encode market structure directly into the seal iconography.

11.4 Scale-Out: Trade Logistics and the Gridiron across the Harappan Civilization

At the Early to Mature transition, it seems like Harappan urbanism just scaled out. The merchants and traders were not trying to scale out—it just looks that way in the archaeological record. They seem to have been singularly focused on just trying to remove any friction that impedes trade.

Possehl talks about Indus ideology, invoking European revolutionary era nihilism as the driving ideology. His assertion seems to be grounded in two particular facts: one, that there seem to have been conflagrations in certain sets of sites, and the fact that "there is a proclivity for Indus sites to be founded on virgin soil; 755 out of 1058 sites, or 71 percent. Of the 523 Early Harappan sites we know of, 324 (62 percent) were abandoned prior to Mature Harappan" (Possehl, 2002).

We posit that the scaling of trade across the Harappan region is a parsimonious explanation. The archaeological record shows toy carts and flat-bottomed boats increasing as the Early Harappan turns Mature, and the humble ox cart has been under-appreciated as a driver of the earliest urban gridiron.

The logistics of this network plausibly grew out of pastoralism. Pastoral groups already moved herds between uplands, plains and river valleys, and so knew the seasonal routes, the water and fodder stops, and the people at either end. As trade developed, some of them plausibly became intermediaries specializing in long-distance movement, first with pack animals and then with carts. Each step added a skill: managing teams of oxen and maintaining carts, choosing and keeping routes, and knowing what was scarce and what was in surplus in distant regions. This last skill, information arbitrage, made them more than carriers. An intermediary who saw both ends of a route could coordinate shipments across the network, lower transaction costs, and let strangers trade with confidence. These are the operators the seal system and the AP2PBS protocol would have served.

Cart geometry—axle width, turning radius, load capacity—set requirements that the organic, winding lanes built for foot and donkey traffic could not meet. Main streets had to let two paired-ox carts pass each other beside pedestrians, and the rounded building corners found at some sites are consistent with protecting walls from passing carts. The gridiron was the capillary system of the cart network, feeding the arteries of the Sindhu and the sea: straight streets reduced friction and perpendicular intersections simplified routing. Carts may thus have accelerated gridiron urbanism, in addition to the light advantages of a north–south alignment.

We propose that the gridiron was laid down when trade scaled, and that it came first as a cart track outside the regular settlement where carts lined up to load and unload, and not as a town plan. Carts needed a broad, straight track just outside the older settlement, where loading, marshalling and storage were convenient. Merchants who sited themselves along the track gained access to the trade, and competition drew others to it, so that within perhaps a generation the settlement's centre of gravity moved to the gridiron. The old winding settlement was then left, burned or built over. Possehl noted this pattern, and there is emerging evidence for this pattern. This is the dynamic we offer against Possehl's "nihilism": the abandonment of many Early Harappan sites and the founding of new ones on virgin soil need not reflect an ideology of rupture, since a shift of the settlement to its new trade axis

would produce the same record. Perhaps the citadels are sites that were built over later. The lower gridiron-layout town is where wealth in a barter economy became concentrated, and urbanism seems to have followed the gradient of increasing trade.

The Early-to-Mature Harappan transition looks in the archaeological record like a phase transition—not gradual linear growth but a relatively rapid nonlinear jump to a new organizational level. That's exactly what growth and preferential attachment network dynamics produce. Once a few nodes adopt the fast (for the Bronze age) logistics-optimized model successfully and begin attracting disproportionate trade traffic, the competitive pressure on remaining non-optimized sites becomes intense and the cascade is fast. The Mature-to-Late Harappan looks like the reverse phase transition—something hit the hubs or top guilds that administered ULC very hard. Late Harappan seems to go back to geometric seal motifs, with local, perhaps family-based trade continuing. Perhaps the local, family-based trade always existed through the Mature Harappan alongside the high-throughput trade across vast distances.

The Mature Harappan package—grid cities, standardized weights, seals, carts, flat-bottomed and maritime boats—isn't a collection of independent innovations that accelerated trade. It's a single integrated response to trade network scaling, appearing together because the competitive pressure from trade scaling that drove one drove all of them simultaneously.

This mirrors modern infrastructure revolutions (arrival of railroads - reorganized cities, inter-state highways - the suburban sprawl, internet - information networks): slow build-up, then rapid reorganization into a new scaling regime, followed by eventual transformation when conditions change.

The absence of palaces, war, and religion, the ubiquitous Unicorn, which we have demonstrated to be aligned with commercial and trade networks, provides evidence fitting a decentralized, economically driven complex commercial network rather than a command hierarchy.

We position Harappan urbanism as a classic example of how complex societies can emerge, thrive, and transform without rulers or priests. The driver of the Mature Harappan was the rapid scaling of the trade network and a civilizational outlook oriented to the reduction of any impediments to trade.

11.5 Periodic Assembly as Standardization Engine

The remarkable uniformity of Harappan standards—brick ratios, weight stones, street grids, seal formats—across 1,500+ sites with scarce evidence of central coercion is explained by the periodic meetings held at the hub sites. The expertise related to all the features that are diagnostic of Harappan civilization would disseminate periodically, the Sindhu template would spread as though by osmosis.

This kind of periodic assembly of a variety of peer expertise perhaps is the most parsimonius explanation of the remarkable standardization over a vast geography. These meetings were driven by the different cycles of the wide array of products and commodities that the Harappans traded - including bricks, sanitation and other technology that is diagnostic of Harappan civilization.

This kind of peer-level periodic meetings was unprecedented historically, perhaps. Never before were there such periodic assembly of expertise at a central places in civilizational level histories during the third millennium BCE.

Voluntary adoption spread because:

- **Demonstration effect:** Hubs like Mohenjo-daro showed operational efficiency of standardization
- **Economic advantage:** Merchants preferentially traded at standardized sites; peripheral sites faced competitive disadvantage
- **Portable expertise:** Master carvers and engineers traveled between sites, disseminating standardization
- **Switching costs:** Once a site adopted standardized bricks and weights, reverting to non-standard materials meant inventory incompatibility

11.6 Scale-Free Robustness Explains 700-Year Longevity

Scale-free networks are robust to random failures but vulnerable to targeted hub disruption (Albert et al., 2000). The Mature Harappan urban phase ($\sim$700 years) exhibited this robustness:

- Loss of small peripheral guilds or sites had minimal network impact
- Merchants could route around failed sites using alternative hub paths
- The AP2PBS protocol itself had no critical single point of failure—any two merchants could authenticate transactions using standardized seals and weights
- Decentralized expert assembly meant no single person or office whose removal could collapse standardization

This explains why the civilization sustained itself across such vast distances and time scales despite lack of centralized authority.

11.7 Hub Collapse and Systemic Cascade Failure

The Late Harappan transition represents network-level cascade failure, not gradual decline. When hub sites failed—whether from environmental stress (Ghaggar-Hakra river shifts), trade disruption (Mesopotamian network collapse), or other factors—the system unraveled suddenly:

- **Loss of standardized weights:** Without hub metrologists maintaining calibration and distributing reference stones, peripheral merchants lost assurance of weight consistency. Trading friction returned.
- **Loss of seal carving expertise:** Without hub artisans maintaining standardized seal templates, new seals could not be commissioned. Merchants reverted to non-standardized, often geometric, seals.

- **Loss of infrastructure coordination:** Without periodic expert assembly, bricks degraded without replacement (architectural standards lost), drainage systems failed, grid-planned settlements became economically unmaintainable.
- **Urbanism dissolution:** Cities could not sustain themselves without the trade surplus that hub coordination generated. Urban depopulation was rapid.
- **Unicorn seal disappearance:** Hub guild production ceased. Peripheral guilds, unprepared for independent seal manufacturing, abandoned the standardized system. Unicorn seals vanish from the archaeological record essentially overnight in regional terms.

The archaeological signature is sudden, not gradual: vanishing urbanism, loss of standardized weights, end of brick uniformity, reversion to geometric seals, fragmentation into local trade networks. This is precisely what scale-free network collapse predicts when hubs fail.

11.8 Script Symbols as Operational Notation System

Decipherment programs, such as the Sangam Rebus theory, Proto-Indo-Aryan readings and general linguistic approaches, begin by assuming that **the script primarily encodes language**, and ask which language the signs record. The present work begins instead from the trade context in which the Unicorn Seal System (USS) appears, and asks what problem the seals were solving pertain to trade. We therefore ask what information a Bronze Age long-distance trade protocol would *need to communicate*. Merchants separated by hundreds of kilometers who exchange goods asynchronously must agree beforehand on standard weights and measures, quality grades, ownership authentication, routing, receiving authority and settlement rules; without these, delayed exchange cannot be verified. The signs are therefore read first as logistical categories (commodity, quantity, container, destination, weight class) and not as words. A boat-like sign is a candidate for a vessel class because vessel type is information such a system would need, not because it resembles a boat. The two programs are different research programs, and each can be judged by how well it explains the archaeological observations: brevity, repetition and the absence of personal identification.

The script appears on seals, tablets and sealings. Inscriptions are short (typically 4–5 signs), highly repetitive, and conspicuously lack personal names and place names. These properties are puzzling if the script records speech, but are expected of operational notation. Mukhopadhyay (2023) concluded that the script does not encode anthroponyms or toponyms but "trade and commodity control," and the stand statistics are consistent with her conclusion: identity was carried visually by the Unicorn, the stand and its attribute bundle, so the inscription did not need to name the parties. Identical signs recur across sites, which indicates standardized codes and not local linguistic variation, and metrology signs occur in spatial association with hexahedral weights, which suggests a notational link between the script and standardized measures.

The rare signs fit the same reading. Of roughly 400 distinct signs, 113 occur only once (hapax legomena) and 47 only twice. If a guild introduced a new commodity, it could agree a provisional sign with its trading partner; the sign would enter the standard repertoire

if trade in the product continued, and would never recur if the transaction was a one-off or the product was discontinued. Singleton stands (representing commercial entities) could likewise generate signs used in isolated transactions. This is a hypothesis consistent with the data, not a demonstration, but it accounts for the long tail of rare signs without assuming incomplete transmission of unknown words.

The three elements of a Unicorn seal—the animal, the stand and the inscription—were rarely manufactured separately, yet decipherment studies usually treat the script alone and glyptic studies the iconography alone. Across more than 3,700 seals the same package recurs, which suggests a unified institutional technology: the animal supports authentication through visual recognition, the stand identifies the commercial entity (guild, warehouse or merchant house), and the script supplies operational metadata for goods in motion. The package is characteristic of periods 3A and 3B, the height of standardization and interregional exchange. In period 3C script-only tablets appear, but they are scarce and occur mainly in workshop areas, suggesting internal production documentation and not authentication tokens for exchange between parties. The framework can thus be evaluated without identifying individual signs: it must explain why Unicorn seals dominate, why inscriptions are short, why sealings are scarcer than seals, why weights are standardized over so wide an area, and why sign order is so regular. If the institutional model (AP2PBS coordinated through the ULC) explains these observations better than competing models, it gains credibility, and what individual signs denote becomes a secondary hypothesis requiring its own evidence.

11.9 Etymology that traces its origins in Harappa

The Harappan civilization was a network of merchants and guilds operating across vast distances. Their periodic meetings at hub sites were perhaps the **commercial sabhās** where these guilds coordinated trade and their ubiquitous standards, perhaps even the entire Sindhu template. These were likely the institutional precursors to the forms that would later be documented as *Śreṇi*, *Nigama*, *Pūga*, *Gana*, and *Sārtha*—the vocabulary of merchant associations and market bodies that appears in texts from the Mauryan and Sangam eras. If this continuity held, the Harappan sabha was where the earliest guilds and merchant associations were formed.

The *Śreṇi*—a term appearing in Sanskrit and Prakrit texts from the Mauryan era onward, likely preserved from pre-Aryan language substrates of South Asia—denotes both "row" or "line" and, by extension, "merchant guild" or "professional association." The Pillared Hall materializes this concept into operational reality. Traders settled their obligations literally sitting in rows—the *Śreṇi* as both spatial arrangement and corporate body. Even prior to Mature Harappan times, perhaps two row of sealings were tallied by counterparties to net a barter settlement when they met for periodic reconciliation.

The figure administering this ordered settlement was the *Śreṣṭhin*, whose title traces to this coordinative function. The word's preservation across centuries and language families suggests institutional continuity: the commercial practices that generated the term's meaning in the Harappan era persisted, in recognizable form, into the documented guild systems of classical India.

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Data Availability

All code, data, and reproducibility documentation are openly available in the GitHub repository: dmahesh2869/harappan-seal-analysis.

The repository is organized in three layers:

- **Core reproducibility pipeline:** primary scripts, validation checks, and run instructions needed to regenerate the manuscript's main statistical results, tables, and figures.
- **Supplementary analyses:** additional scripts used for robustness checks and exploratory diagnostics (e.g., stand hierarchy, bundle consistency, singleton dispersion, and diachronic analyses).
- **Versioned outputs:** generated CSV, TXT, and PNG artifacts used to audit and verify claims in the manuscript.

All analyses are deterministic and reproducible in Python 3.8+ with `powerlaw` (v2.0.0).

12 Conclusion

This paper has argued that the Harappan Civilization is best understood as a merchant commonwealth. It was a merchant commonwealth organized as a long distance barter trade network—unprecedented in human history and never fully replicated since.

At its operational core lay the what we propose as periodic meetings, the USS, the AP2PBS (Asynchronous Peer-to-Peer Barter Settlement) protocol or something akin to it and the Unicorn Logistics Confederation (ULC)—a confederation of leading merchant guilds—provided the institutional framework. It guaranteed weight standardization, authorized seals, arbitrated disputes, secured delivery, underwrote counterparty risk, and coordinated multimodal logistics. Six functions that made the system work.

This civilization demonstrates how complex societies emerge, thrive, and transform without script (in the linguistic sense), without rulers, and without priests. The driving motive was perhaps simple: reduction of friction to trade. Every innovation served that motive. The USS was iteratively refined to reduce friction and enhance the efficiency of trade. Every architectural form followed that function.

It is poetic that when Louis Sullivan articulated his architectural dictum at the turn of the twentieth century, he was ushering in the age of vertical urbanism—soaring skyscrapers rising toward the sky in response to the density requirements of machine-age cities. Yet the Harappans, more than 4,600 years earlier, had perfected Sullivan's very principle when they organized the horizontal gridiron. Every alley, every wall thickness, every drain, every warehouse orientation followed directly from the functional requirements of long-distance cartage and exchange. They proved that function could be sovereign without abandoning beauty—that the most elegant city might be the most economically efficient one.

Form ever follows function. This is the law. — Louis H. Sullivan

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